

# INTEGRATION

## Standard results

- (1)  $\int x^n dx = \frac{x^{n+1}}{n+1} + C$  if  $n \neq -1$
- (2)  $\int 1 dx = x + C$
- (3)  $\int \frac{1}{x} dx = \log x + C$  (or  $\ln x + C$ )
- (4)  $\int \sqrt{x} dx = \frac{2}{3} x^{3/2} + C$
- (5)  $\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$
- (6)  $\int e^x dx = e^x + C$
- (7)  $\int a^x dx = \frac{a^x}{\log a} + C$  where  $a$  is any constant
- (8)  $\int \sin x dx = -\cos x + C$
- (9)  $\int \cos x dx = \sin x + C$
- (10)  $\int \sec^2 x dx = \tan x + C$
- (11)  $\int \operatorname{cosec}^2 x dx = -\cot x + C$
- (12)  $\int \sec x \cdot \tan x dx = \sec x + C$
- (13)  $\int \operatorname{cosec} x \cdot \cot x dx = -\operatorname{cosec} x + C$
- (14)  $\int \tan x dx = \log(\sec x) + C$  (or  $-\log(\cos x) + C$ )
- (15)  $\int \cot x dx = \log(\sin x) + C$
- (16)  $\int \sec x dx = \log(\sec x + \tan x) + C$
- (17)  $\int \operatorname{cosec} x dx = \log(\operatorname{cosec} x + \cot x) + C$   
or  
 $\log \tan \frac{x}{2} + C$

$$(18) \int (ax+b)^n dx = \frac{1}{a} \cdot \frac{(ax+b)^{n+1}}{n+1} + c \quad \text{if } n \neq -1$$

$$(19) \int \frac{1}{ax+b} dx = \frac{1}{a} \cdot \log(ax+b) + c$$

$$(20) \int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + c$$

$$(21) \int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + c$$

$$(22) \int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + c$$

$$(23) \int (f(x))^n f'(x) dx = \frac{[f(x)]^{n+1}}{n+1} + c ; n \neq -1$$

$$(24) \int \frac{f'(x)}{f(x)} dx = \log f(x) + c$$

$$(25) \int \phi(f(x)) f'(x) dx = \int \phi(t) dt \quad \text{where } t = f(x).$$

Integrate the following functions

(i)  $\sqrt{2x+3}$       (ii)  $\sin^2 x$       (iii)  $\cos^3 x$

(iv)  $\sin 2x \cos 4x$       (v)  $\sin 5x \cdot \sin 2x$

(vi)  $(x^2 + a^2)^3 x$       (vii)  $\sin^7 x \cos x$

(viii)  $\frac{\sin(\log x)}{x}$       (ix)  $\frac{x^3}{(1+x^2)^5}$       (x)  $\frac{1}{\sqrt{3x-2}}$

### Answers

$$\begin{aligned} \text{(i)} \int (2x+3)^{1/2} dx &= \frac{1}{2} \cdot \frac{(2x+3)^{3/2}}{3/2} + c \\ &= \frac{1}{3} (2x+3)^{3/2} + c \end{aligned}$$

$$\begin{aligned} \text{(ii)} \text{ we know that } \sin^2 x &= \frac{1 - \cos 2x}{2} \\ \therefore \int \sin^2 x dx &= \int \frac{1 - \cos 2x}{2} dx = \frac{1}{2} \left[ x - \frac{\sin 2x}{2} \right] + c \end{aligned}$$

(iii) we know that

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

$$\therefore 4 \cos^3 x = \cos 3x + 3 \cos x$$

$$\text{ie } \cos^3 x = \frac{1}{4} (\cos 3x + 3 \cos x)$$

$$\begin{aligned} \int \cos^3 x \, dx &= \frac{1}{4} \int (\cos 3x + 3 \cos x) \, dx \\ &= \frac{1}{4} \left[ \frac{\sin 3x}{3} + 3 \sin x \right] + C \end{aligned}$$

(iv) we know that

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$\begin{aligned} \therefore \sin 2x \cos 4x &= \frac{1}{2} (2 \sin 2x \cos 4x) \\ &= \frac{1}{2} [\sin 6x + \sin(-2x)] \\ &= \frac{1}{2} [\sin 6x - \sin 2x] \end{aligned}$$

$$\begin{aligned} \therefore \int \sin 2x \cos 4x \, dx &= \frac{1}{2} \int (\sin 6x - \sin 2x) \, dx \\ &= \frac{1}{2} \left[ -\frac{\cos 6x}{6} + \frac{\cos 2x}{2} \right] + C \end{aligned}$$

(v) we know that

$$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$\begin{aligned} \therefore \sin 5x \sin 2x &= \frac{1}{2} (2 \sin 5x \sin 2x) \\ &= \frac{1}{2} (\cos(3x) - \cos 7x) \end{aligned}$$

$$\begin{aligned} \int \sin 5x \sin 2x \, dx &= \frac{1}{2} \int (\cos 3x - \cos 7x) \, dx \\ &= \frac{1}{2} \left[ \frac{\sin 3x}{3} - \frac{\sin 7x}{7} \right] + C \end{aligned}$$

$$(vi) \int (x^2 + a^2)^3 x \, dx = \int u^3 \frac{du}{2}$$

$$= \frac{1}{2} \frac{u^4}{4} + C$$

$$= \frac{(x^2 + a^2)^4}{8} + C$$

$$\begin{cases} \text{Put } u = x^2 + a^2 \\ du = 2x \, dx \\ \frac{du}{2} = x \, dx \end{cases}$$

$$\begin{aligned}
 \text{(vii)} \quad \int \sin^7 x \cos x \, dx &= \int u^6 du \quad \left( \begin{array}{l} u = \sin x \\ du = \cos x \, dx \end{array} \right) \\
 &= \frac{u^7}{7} + C \\
 &= \frac{\sin^7 x}{7} + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(viii)} \quad \int \frac{\sin(\log x)}{x} \, dx &= \int \sin(\log x) \cdot \frac{1}{x} \, dx \quad \left( \begin{array}{l} \text{Put } u = \log x \\ du = \frac{1}{x} \, dx \end{array} \right) \\
 &= \int \sin u \cdot du \\
 &= -\cos u + C \\
 &= -\cos(\log x) + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(ix)} \quad \int \frac{x^3}{(1+x^2)^5} \, dx &= \int \frac{x^2}{(1+x^2)^5} x \, dx \\
 &= \int \frac{u-1}{u^5} \frac{du}{2} \\
 &= \frac{1}{2} \int \left( \frac{1}{u^4} - \frac{1}{u^5} \right) du \\
 &= \frac{1}{2} \int (u^{-4} - u^{-5}) \, du \\
 &= \frac{1}{2} \left[ \frac{u^{-3}}{-3} - \frac{u^{-4}}{-4} \right] + C \\
 &= \frac{1}{2} \left[ -\frac{1}{3u^3} + \frac{1}{4u^4} \right] + C \\
 &= \frac{1}{2u^4} \left[ -\frac{u}{3} + \frac{1}{4} \right] + C \\
 &= \frac{1}{2(1+x^2)^4} \left[ -\frac{(1+x^2)}{3} + \frac{1}{4} \right] + C
 \end{aligned}$$

Put  $u = 1+x^2$   
 $du = 2x \, dx$   
 $x \, dx = \frac{du}{2}$   
 Also  $x^2 = u-1$

$$\begin{aligned}
 \text{(x)} \quad \int \frac{1}{\sqrt{3x-2}} \, dx &= \int (3x-2)^{-1/2} \, dx \\
 &= \frac{1}{3} \frac{(3x-2)^{1/2}}{1/2} + C \\
 &= \frac{2}{3} \sqrt{3x-2} + C
 \end{aligned}$$



## Some more standard results

- (1)  $\int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$
- (2)  $\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$
- (3)  $\int \frac{dx}{a^2-x^2} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C$
- (4)  $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$
- (5)  $\int \frac{dx}{\sqrt{x^2+a^2}} = \log(x + \sqrt{x^2+a^2}) + C$
- (6)  $\int \frac{dx}{\sqrt{x^2-a^2}} = \log(x + \sqrt{x^2-a^2}) + C$
- (7)  $\int \sqrt{a^2-x^2} dx = \frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + C$
- (8)  $\int \sqrt{x^2+a^2} dx = \frac{x}{2} \sqrt{x^2+a^2} + \frac{a^2}{2} \log(x + \sqrt{x^2+a^2}) + C$
- (9)  $\int \sqrt{x^2-a^2} dx = \frac{x}{2} \sqrt{x^2-a^2} - \frac{a^2}{2} \log(x + \sqrt{x^2-a^2}) + C$

Integrate the following

- (i)  $\frac{1}{x^2+2x+2}$       (ii)  $\frac{1}{25x^2-16}$       (iii)  $\frac{1}{\sqrt{81+64x^2}}$
- (iv)  $\frac{1}{\sqrt{8-2x-x^2}}$       (v)  $\sqrt{4-x^2}$

## Answers

$$\begin{aligned}
 \text{(i)} \quad \int \frac{dx}{x^2+2x+2} &= \int \frac{dx}{(x+1)^2+1^2} \\
 &= \frac{1}{1} \tan^{-1}\left(\frac{x+1}{1}\right) + C \\
 &= \tan^{-1}(x+1) + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \int \frac{dx}{25x^2-16} &= \int \frac{dx}{25\left(x^2-\frac{16}{25}\right)} \\
 &= \frac{1}{25} \int \frac{dx}{x^2-\left(\frac{4}{5}\right)^2}
 \end{aligned}$$

$$= \frac{1}{25} \cdot \frac{1}{2 \times \frac{4}{5}} \log \left| \frac{x - \frac{4}{5}}{x + \frac{4}{5}} \right| + C$$

$$= \frac{1}{40} \log \left| \frac{5x-4}{5x+4} \right| + C$$

$$(iii) \int \frac{1}{\sqrt{81+64x^2}} dx = \int \frac{dx}{\sqrt{64 \left( \frac{81}{64} + x^2 \right)}}$$

$$= \frac{1}{8} \int \frac{dx}{\sqrt{\left( \frac{9}{8} \right)^2 + x^2}}$$

$$= \frac{1}{8} \log \left[ x + \sqrt{\left( \frac{9}{8} \right)^2 + x^2} \right] + C$$

$$= \frac{1}{8} \log \left[ x + \sqrt{\frac{81}{64} + x^2} \right] + C$$

$$(iv) \int \frac{dx}{\sqrt{8-2x-x^2}} = \int \frac{dx}{\sqrt{8-(x^2+2x)}}$$

$$= \int \frac{dx}{\sqrt{9-(x^2+2x+1)}}$$

$$= \int \frac{dx}{\sqrt{3^2-(x+1)^2}}$$

$$= \sin^{-1} \left( \frac{x+1}{3} \right) + C$$

$$(v) \int \sqrt{4-x^2} dx = \int \sqrt{2^2-x^2} dx$$

$$= \frac{x}{2} \sqrt{2^2-x^2} + \frac{2^2}{2} \sin^{-1} \left( \frac{x}{2} \right) + C$$

$$= \frac{x}{2} \sqrt{4-x^2} + 2 \sin^{-1} \left( \frac{x}{2} \right) + C$$

### Assignment problems

Integrate

(i)  $\sqrt{3-2x}$

(ii)  $\cos^2 x$

(iii)  $\sin^3 x$

(iv)  $\cos 5x \cdot \cos x$

(v)  $x^2 \cos(x^3+1)$

(vi)  $e^{\tan x} \sec^2 x$

(vii)  $\frac{1}{\sqrt{x^2+3x+1}}$

(viii)  $\frac{1}{x^2+4x+13}$

(ix)  $\sqrt{9-36x^2}$

## Integration by parts

This is a method to integrate product of two functions.

$$\int uv dx = u \int v dx - \int \left( \frac{du}{dx} \int v dx \right) dx$$

$$\text{or } \int \text{I} \cdot \text{II} = \text{I} \int \text{II} - \int [d(\text{I}) \int \text{II}]$$

Integrate the following

(i)  $x \cos x$       (ii)  $x^2 \sin x$       (iii)  $x \log x$

(iv)  $\tan^{-1} x$       (v)  $x^2 e^{3x}$

$$\begin{aligned} \text{(i)} \quad \int \underset{\text{I}}{x} \underset{\text{II}}{\cos x} dx &= \text{I} \int \text{II} - \int (d(\text{I}) \int \text{II}) \\ &= x \sin x - \int 1 \cdot \sin x dx \\ &= x \sin x - (-\cos x) + C \\ &= x \sin x + \cos x + C \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \int \underset{\text{I}}{x^2} \underset{\text{II}}{\sin x} dx &= x^2 (-\cos x) - \int 2x (-\cos x) dx \\ &= -x^2 \cos x + 2 \int \underset{\text{I}}{x} \underset{\text{II}}{\cos x} dx \\ &= -x^2 \cos x + 2 \left[ x \sin x - \int 1 \cdot \sin x dx \right] \\ &= -x^2 \cos x + 2 \{ x \sin x - (-\cos x) \} + C \\ &= (2 - x^2) \cos x + 2x \sin x + C \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \int \underset{\text{II}}{x \log x} \underset{\text{I}}{dx} &= \log x \cdot \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} dx \\ &= \frac{x^2}{2} \log x - \frac{1}{2} \int x dx \\ &= \frac{x^2}{2} \log x - \frac{1}{2} \left( \frac{x^2}{2} \right) + C \\ &= \frac{x^2}{2} \left( \log x - \frac{1}{2} \right) + C \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad \int \underset{\text{I}}{\tan^{-1} x} \underset{\text{II}}{dx} &= \int \tan^{-1} x \cdot \frac{1}{x} dx \\ &= \tan^{-1} x \cdot x - \int \frac{1}{1+x^2} x dx \end{aligned}$$

$$= x + \tan^{-1} x - \frac{1}{2} \int \frac{2x}{1+x^2} dx$$

$$= x + \tan^{-1} x - \frac{1}{2} \log(1+x^2) + C$$

$$\begin{aligned} \text{(v)} \int \underbrace{x^2}_{\text{I}} \underbrace{e^{3x}}_{\text{II}} dx &= \frac{x^2 e^{3x}}{3} - \int 2x \frac{e^{3x}}{3} dx \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \int \frac{x e^{3x}}{\text{I} \text{ II}} dx \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left[ x \frac{e^{3x}}{3} - \int 1 \cdot \frac{e^{3x}}{3} dx \right] \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left[ \frac{x e^{3x}}{3} - \frac{1}{3} \frac{e^{3x}}{3} \right] + C \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{9} x e^{3x} + \frac{2}{27} e^{3x} \\ &= \frac{e^{3x}}{27} [9x^2 - 6x + 2] \end{aligned}$$

short cut method

$$\int u v dx = u v_1 - u' v_2 + u'' v_3 - \dots$$

where  $u' = \frac{du}{dx}$ ,  $u'' = \frac{d^2 u}{dx^2}$ ;  $v_1 = \int v dx$ ,  $v_2 = \int v_1 dx$  ...

$$\begin{aligned} \text{(i)} \int x \cos x dx &= x \sin x - 1 \cdot (-\cos x) + C \quad \begin{pmatrix} u=x \\ v=\cos x \end{pmatrix} \\ &= x \sin x + \cos x + C \end{aligned}$$

$$\begin{aligned} \text{(ii)} \int \underbrace{x^2}_u \underbrace{\sin x}_v dx &= x^2 (-\cos x) - 2x (-\sin x) + 2 \cdot (\cos x) + C \\ &= (2-x^2) \cos x + 2x \sin x + C \end{aligned}$$

$$\begin{aligned} \text{(iii)} \int \underbrace{x^2}_u \underbrace{e^{3x}}_v dx &= x^2 \cdot \frac{e^{3x}}{3} - 2x \frac{e^{3x}}{9} + 2 \cdot \frac{e^{3x}}{27} + C \\ &= \frac{e^{3x}}{27} [9x^2 - 6x + 2] + C \end{aligned}$$

Integrate

(i)  $x \sin x$

(iii)  $x^3 e^{2x}$

(v)  $\sin^{-1} x$

(ii)  $x^2 \cos x$

(iv)  $\log x$

Two more results

$$(1) \int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + C$$

$$(2) \int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + C$$

Evaluate (i)  $\int e^{2x} \sin 3x \, dx$  (ii)  $\int e^{3x} \cos x \, dx$

(i)  $a=2, b=3$

$$\begin{aligned} \therefore \int e^{2x} \sin 3x \, dx &= \frac{e^{2x}}{2^2 + 3^2} (2 \sin 3x - 3 \cos 3x) + C \\ &= \frac{e^{2x}}{13} (2 \sin 3x - 3 \cos 3x) + C \end{aligned}$$

(ii)  $a=3, b=1$

$$\begin{aligned} \therefore \int e^{3x} \cos x \, dx &= \frac{e^{3x}}{3^2 + 1^2} (3 \cos x + 1 \cdot \sin x) + C \\ &= \frac{e^{3x}}{10} (3 \cos x + \sin x) + C \end{aligned}$$

Integrate (i)  $e^{3x} \sin x$  (ii)  $e^x \cos 2x$



## DIFFERENTIAL EQUATIONS

An equation in which differential coefficients occur is called a differential equation

### Examples

$$(1) (2x+3y) \frac{dy}{dx} + (3x-4)y = 0$$

$$(2) \frac{d^2 y}{dx^2} + y = 0$$

$$(3) \frac{d^2 y}{dx^2} - \left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2} = 0$$

$$(4) \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$$

$$(5) \left(\frac{d^3 y}{dx^3}\right)^2 - 3x \left(\frac{dy}{dx}\right)^5 + 5y = 0$$

$$(6) x \frac{\partial^2 z}{\partial x^2} - y \frac{\partial^2 z}{\partial y^2} = x^2 - y^2$$

### Ordinary differential equations

A differential equation involving a single independent variable and hence ordinary derivatives is called an ordinary diff. eq

Examples (1), (2), (3) and (5) are ordinary diff. eqs.

### Partial differential equations

If there are two or more independent variables present, so that the equation contains partial derivatives, it is called a partial diff. eq.

Examples (4) and (6) are partial diff. eqs.

Order of a diff. eq      order of a diff. eq

is the order of the highest order derivative occurring in it.

Degree of a diff. eq.

degree of a diff. eq

is the degree of the highest order derivative occurring in it.

Example (2) first order first degree

eq (2) order 2 degree 1

eq (3) can be written as

$$\frac{d^2 y}{dx^2} = \left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2} \text{ squaring } \left(\frac{d^2 y}{dx^2}\right)^2 = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^3$$

Now fractional power is removed

$\therefore$  It is of order 2 and degree 2

eq (4) order-2 and degree 1 partial diff. eq

eq (5) order-3 degree 2

eq (6) is a first order first degree partial diff. eq.

### Solution of a diff. eq

A relation between the variables not involving derivatives which satisfy the given diff. eq is called a solution of the diff. eq. A general solution of a diff. eq contains as many arbitrary constants as order of the eq. Giving particular values to arbitrary constants we get particular solutions.

Example consider the diff. eq.  $\frac{dy}{dx} - x = 0$

$$\text{i.e. } \frac{dy}{dx} = x \Rightarrow dy = x dx$$

$$\text{Integrating } \int dy = \int x dx$$

$$\text{i.e. } y = \frac{x^2}{2} + \frac{c}{2} \Rightarrow 2y = x^2 + c$$

is the general solution

putting  $c=0$ , we get the particular soln  $2y=x^2$ , and putting  $c=1$ , we get the particular solution  $2y=x^2+1$

Note consider the relation  $2y=x^2+c$

diff. w.r. to  $x$ ,  $2 \frac{dy}{dx} = 2x \Rightarrow \frac{dy}{dx} = x$

i.e.  $\frac{dy}{dx} - x = 0$  which is the diff. eq corresponding to the above relation

## FIRST ORDER FIRST DEGREE EQUATIONS

### Type I Variable separable equations

A diff. eq of the form  $M(x,y)dx + N(x,y)dy = 0$  is a variable separable eq. if it can be expressed with the form  $f(x)dx + g(y)dy = 0$

Solve (1)  $\frac{dy}{dx} = \frac{y}{x}$  (2)  $(y+2)dx + y(x+4)dy = 0$

$$(3) x \sin y dx + (x^2+1) \cos y dy = 0$$

$$(4) 4xy dx + (x^2+1) dy = 0$$

$$(5) \tan \theta dr + 2r d\theta = 0$$

$$(1) \frac{dy}{dx} = \frac{y}{x} \Rightarrow \frac{dy}{y} = \frac{dx}{x}$$

$$\text{Integrating, } \int \frac{dy}{y} = \int \frac{dx}{x} \Rightarrow \log y = \log x + \log c$$

$$\text{i.e. } \log y = \log(cx) \Rightarrow y = cx \text{ is the general}$$

solution.

$$(2) (y+2)dx + y(x+4)dy = 0$$

Dividing by:  $(x+4)(y+2)$

$$\frac{dx}{x+4} + \frac{y}{y+2} dy = 0 \Rightarrow \frac{dx}{x+4} + \frac{y+2-2}{y+2} dy = 0$$

$$\text{i.e. } \frac{dx}{x+4} + \left(1 - \frac{2}{y+2}\right) dy = 0$$



Integrating  $\log(x+4) + y - 2\log(y+2) = C$   
is the general solution

$$(3) \quad x \sin y \, dx + (x^2+1) \cos y \, dy = 0$$

Dividing by  $(x^2+1) \sin y$

$$\frac{x}{x^2+1} dx + \frac{\cos y}{\sin y} dy = 0$$

$$\therefore \int \frac{x}{x^2+1} dx + \int \frac{\cos y}{\sin y} dy = 0$$

$$\text{Multiplying by 2, } \int \frac{2x}{x^2+1} dx + 2 \int \frac{\cos y}{\sin y} dy = 0$$

$$\therefore \log(x^2+1) + 2\log \sin y = \log C$$

$$\therefore \log(x^2+1) \sin^2 y = \log C \Rightarrow (x^2+1) \sin^2 y = C$$

$$(4) \quad 4xy \, dx + (x^2+1) dy = 0$$

Dividing by  $(x^2+1)y$ ,

$$\frac{4x \, dx}{x^2+1} + \frac{dy}{y} = 0$$

$$\text{Integrating } \int \frac{4x}{x^2+1} dx + \int \frac{dy}{y} = 0$$

$$\therefore 2 \int \frac{2x}{x^2+1} dx + \int \frac{dy}{y} = 0$$

$$\therefore 2\log(x^2+1) + \log y = \log C$$

$$\log(x^2+1)^2 y = \log C$$

soln is given by  $(x^2+1)^2 y = C$

$$(5) \quad r \tan \theta \, dr + 2r \, d\theta$$

$$\text{Dividing } r \tan \theta, \quad \frac{dr}{r} + \frac{2}{\tan \theta} d\theta = 0$$

$$\Rightarrow \int \frac{dr}{r} + 2 \int \cot \theta \, d\theta = 0$$

$$\Rightarrow \log r + 2\log \sin \theta = \log C$$

$$\Rightarrow \log r \sin^2 \theta = \log C$$

$\therefore r \sin^2 \theta = C$  is the solution.

## Homogeneous equations

A diff. eq which can be reduced to the form  $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$  is called a homogeneous diff. eq. This eq can be solved by putting  $y = vx$  and hence reducing it to variable separable form

Solve the following diff. eqs

$$(1) \quad 2xy \frac{dy}{dx} - y^2 + x^2 = 0$$

$$(2) \quad xy' = x + y$$

$$(3) \quad \frac{dy}{dx} = \frac{y+x}{y-x}$$

$$(4) \quad (y + \sqrt{x^2 + y^2})dx - xdy = 0$$

$$(1) \quad 2xy \frac{dy}{dx} = y^2 - x^2 \Rightarrow \frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 \left( \frac{y^2}{x^2} - 1 \right)}{2xy} \Rightarrow \frac{dy}{dx} = \frac{\left( \frac{y}{x} \right)^2 - 1}{2 \left( \frac{y}{x} \right)} \quad -xv$$

$\therefore$  It is a homogeneous diff. eq

$$\text{Put } y = vx \quad \therefore \frac{dy}{dx} = v + x \cdot \frac{dv}{dx}$$

$$\text{Sub in eq } v + x \frac{dv}{dx} = \frac{v^2 - 1}{2v}$$

$$\therefore x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v = \frac{v^2 - 1 - 2v^2}{2v}$$

$$\therefore x \frac{dv}{dx} = -\frac{(v^2 + 1)}{2v} \Rightarrow \frac{2v}{v^2 + 1} dv = -\frac{dx}{x}$$

$$\text{Integrating, } \int \frac{2v}{v^2 + 1} dv + \int \frac{dx}{x} = 0$$

$$\text{ie } \log(v^2 + 1) + \log x = \log c$$

$$\text{ie } \log(v^2 + 1)x = \log c$$

$$\text{ie } (v^2 + 1)x = c$$

$$\text{ie } \left( \frac{y^2}{x^2} + 1 \right) x = c$$

$$\text{ie } \frac{y^2 + x^2}{x^2} \cdot x = c \Rightarrow \underline{\underline{x^2 + y^2 = cx}}$$

$$( \because v = \frac{y}{x} )$$



$$(2) \quad xy' = x + y \Rightarrow \frac{dy}{dx} = \frac{x+y}{x} \rightarrow (1)$$

$$\Rightarrow \frac{dy}{dx} = 1 + \frac{y}{x}$$

$$\text{Put } y = vx \quad \therefore \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\text{Eq (1) reduces to } v + x \frac{dv}{dx} = 1 + v$$

$$\text{or } x \frac{dv}{dx} = 1 \Rightarrow dv = \frac{dx}{x}$$

$$\text{Integrating } v = \log x + \log c$$

$$\text{or } v = \log cx \Rightarrow \frac{y}{x} = \log cx$$

$\therefore y = x \log cx$  is the solution.

$$(3) \quad \frac{dy}{dx} = \frac{y+x}{y-x} \Rightarrow \frac{dy}{dx} = \frac{x(\frac{y}{x}+1)}{x(\frac{y}{x}-1)} = \frac{\frac{y}{x}+1}{\frac{y}{x}-1}$$

$$\text{put } y = vx, \quad \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\therefore \text{ we get } v + x \frac{dv}{dx} = \frac{v+1}{v-1}$$

$$\therefore x \frac{dv}{dx} = \frac{v+1}{v-1} - v \Rightarrow x \frac{dv}{dx} = \frac{v+1-(v^2-v)}{v-1}$$

$$\text{or } x \frac{dv}{dx} = \frac{-v^2+2v+1}{v-1} \Rightarrow x \frac{dv}{dx} = -\frac{(v^2-2v-1)}{v-1}$$

$$\Rightarrow \frac{v-1}{v^2-2v-1} dv = -\frac{dx}{x}$$

$$\text{Multiplying by 2, } \frac{2v-2}{v^2-2v-1} dv + 2 \frac{dx}{x} = 0$$

$$\Rightarrow \int \frac{2v-2}{v^2-2v-1} dv + 2 \int \frac{dx}{x} = 0$$

$$\Rightarrow \log(v^2-2v-1) + 2 \log x = \log c$$

$$\Rightarrow \log(v^2-2v-1)x^2 = \log c \Rightarrow (v^2-2v-1)x^2 = c$$

$$\Rightarrow \left(\frac{y^2}{x^2} - 2\frac{y}{x} - 1\right)x^2 = c \Rightarrow y^2 - 2xy - x^2 = c$$

$$(4) \quad \frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x} \quad \text{Put } y = vx$$

$$x \frac{dv}{dx} = \sqrt{1+v^2}$$

$$\text{soln } y + \sqrt{x^2 + y^2} = cx^2$$

## Linear diff. eq

A diff. eq is said to be linear when the dependent variable and its derivatives occur only in first degree and are not multiplied together.

Eg (1)  $x \frac{dy}{dx} + y = 1$  (2)  $\frac{dy}{dx} - y = e^{2x}$

The diff. eq  $y \frac{dy}{dx} + x = 1$  is not a linear diff. eq

General form of a first order linear diff. eq

is  $\frac{dy}{dx} + P(x)y = Q(x)$

### Method of finding solution

$e^{\int P(x) dx}$  is called the integrating factor  
(I.F). Soln of the above is given by

$$y e^{\int P(x) dx} = \int (e^{\int P(x) dx} \cdot Q(x)) dx$$

$$\text{i.e. } y (I.F) = \int (I.F) Q(x) dx$$

Solve the following diff. eq

(1)  $\frac{dy}{dx} - y = e^{2x}$  (2)  $(x^2+1) \frac{dy}{dx} + 4xy = x$

(3)  $\frac{dy}{dx} + y \tan x = \cos^3 x$  (4)  $\frac{dy}{dx} - y \tan x = e^{2x} \sec x$

### Solutions

(1)  $\frac{dy}{dx} - y = e^{2x}$   $P = -1, Q = e^{2x}$

$$I.F. = e^{\int P dx} = e^{\int (-1) dx} = e^{-x}$$

$$\begin{aligned} \text{soln, } y(e^{-x}) &= \int e^{-x} \cdot e^{2x} \cdot dx \\ &= \int e^x dx \\ &= e^x + c \end{aligned}$$

$$\therefore y = e^x (e^x + c)$$

$$(ii) (x^2+1) \frac{dy}{dx} + 4xy = x$$

Dividing by  $(x^2+1)$ , we get

$$\frac{dy}{dx} + \frac{4x}{x^2+1} y = \frac{x}{x^2+1}$$

$$P = \frac{4x}{x^2+1}, \quad Q = \frac{x}{x^2+1}$$

$$\begin{aligned} I.F. &= e^{\int \frac{4x}{x^2+1} dx} = e^{2 \int \frac{2x}{x^2+1} dx} = e^{2 \log(x^2+1)} \\ &= e^{\log(x^2+1)^2} = (x^2+1)^2 \end{aligned}$$

$$\text{Solution: } y(x^2+1)^2 = \int (x^2+1)^2 \cdot \frac{x}{x^2+1} dx$$

$$= \int (x^2+1) x dx$$

$$= \int (x^3 + x) dx$$

$$\therefore y(x^2+1)^2 = \frac{x^4}{4} + \frac{x^2}{2} + C \quad \text{is the soln.}$$

$$(3) \frac{dy}{dx} + y \tan x = \cos^3 x$$

$$P = \tan x, \quad Q = \cos^3 x$$

$$I.F. = e^{\int \tan x dx} = e^{\log \sec x} = \sec x$$

$$\text{Soln: } y(IF) = \int (IF) Q dx$$

$$\therefore y \sec x = \int \sec x \cdot \cos^3 x dx$$

$$= \int \cos^2 x dx$$

$$= \int \frac{1 + \cos 2x}{2} dx$$

$$= \frac{1}{2} \left[ x + \frac{\sin 2x}{2} \right] + C$$

$$\therefore y \sec x = \frac{1}{4} [2x + \sin 2x] + C$$

$$(4) P = -\tan x, \quad Q = e^x \cdot \sec x$$

$$I.F. = \cos x$$

$$\text{Soln: } y \cos x = e^x + C$$

Bernoulli's differential equation

A diff. eq. of the form

$\frac{dy}{dx} + P(x)y = Q(x)y^n$  where  $n$  is a real number except 0 and 1 is called a Bernoulli's diff. eq.

Method Dividing the eq by  $y^n$

$$\frac{1}{y^n} \frac{dy}{dx} + P(x) \frac{1}{y^{n-1}} = Q(x)$$

$$\text{i.e. } y^{-n} \frac{dy}{dx} + P(x) y^{1-n} = Q(x) \rightarrow (1)$$

$$\text{Put } z = y^{1-n} \quad \therefore \frac{dz}{dx} = (1-n) y^{-n} \frac{dy}{dx}$$

$$\text{i.e. } \frac{1}{1-n} \frac{dz}{dx} = y^{-n} \frac{dy}{dx}$$

$$\text{sub. in eq (1), } \frac{1}{1-n} \frac{dz}{dx} + P(x)z = Q(x)$$

$$\text{i.e. } \frac{dz}{dx} + (1-n)P(x)z = (1-n)Q(x)$$

This is a linear diff. eq. with  $z$  as dependent variable.

Solve the following diff. eqs.

$$(1) \frac{dy}{dx} + 2y = y^2$$

$$(2) x \frac{dy}{dx} + y = xy^3$$

$$(3) \frac{dy}{dx} - y \tan x = y^4 \sec x$$

$$(4) \frac{dy}{dx} + \tan x \cdot \tan y = \cos x \cdot \sec y$$

$$(5) \frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$$

Solutions (1)  $\frac{dy}{dx} + 2y = y^2$

Dividing by  $y^2$ ,  $\frac{1}{y^2} \frac{dy}{dx} + \frac{2}{y} = 1$

$$\text{i.e. } y^{-2} \frac{dy}{dx} + 2y^{-1} = 1 \rightarrow (1)$$

$$\text{Put } z = y^{-1}, \quad \frac{dz}{dx} = -1 y^{-2} \frac{dy}{dx}, \quad -\frac{dz}{dx} = y^{-2} \frac{dy}{dx}$$



Sub in (1)  $-\frac{dz}{dx} + 2z = 1$

i.e.  $\frac{dz}{dx} - 2z = -1$  This is a linear diff eq.

$P = -2, Q = -1. \therefore I.F = e^{-2x}$

Solution  $y(I.F) = \int (I.F) Q dx$

i.e.  $z e^{-2x} = \int -e^{-2x} dx$   
 $= -\frac{e^{-2x}}{-2} + \frac{C}{2}$

i.e.  $2z e^{-2x} = e^{-2x} + C$   
 $\frac{2}{y} e^{-2x} = e^{-2x} + C$

i.e.  $\frac{2}{y} = 1 + C e^{2x}$  is the solution.

(2)  $x \frac{dy}{dx} + y = x y^3$

Dividing by  $x y^3$

$\frac{1}{y^3} \frac{dy}{dx} + \frac{1}{x y^2} = 1$  i.e.  $y^{-3} \frac{dy}{dx} + \frac{1}{x} y^{-2} = 1 \rightarrow (1)$

Put  $z = y^{-2} \Rightarrow \frac{dz}{dx} = -2 y^{-3} \frac{dy}{dx} \Rightarrow -\frac{1}{2} \frac{dz}{dx} = y^{-3} \frac{dy}{dx}$

$\therefore$  we get  $-\frac{1}{2} \frac{dz}{dx} + \frac{1}{x} z = 1$

i.e.  $\frac{dz}{dx} - \frac{2}{x} z = -2, P = -\frac{2}{x}, Q = -2$

$I.F = e^{\int -\frac{2}{x} dx} = e^{-2 \log x} = e^{\log x^{-2}} = \frac{1}{x^2}$

Solution:  $z(I.F) = \int (I.F) Q dx$

i.e.  $z \cdot \frac{1}{x^2} = \int \frac{1}{x^2} (-2) dx$

i.e.  $z \cdot \frac{1}{x^2} = 2 \int -\frac{1}{x^2} dx$   
 $= 2 \left( \frac{1}{x} \right) + C$

i.e.  $z = 2x + C x^2$

i.e.  $\frac{1}{y^2} = 2x + C x^2$  is the solution.

$$(3) \frac{dy}{dx} - y \tan x = y^4 \sec x$$

Dividing by  $y^4$

$$\frac{1}{y^4} \frac{dy}{dx} - \frac{1}{y^3} \tan x = \sec x$$

$$\text{or } y^{-4} \frac{dy}{dx} - y^{-3} \tan x = \sec x$$

$$\text{put } z = y^{-3}, \quad \frac{dz}{dx} = -3y^{-4} \frac{dy}{dx}, \quad -\frac{1}{3} \frac{dz}{dx} = y^{-4} \frac{dy}{dx}$$

$$\text{or } -\frac{1}{3} \frac{dz}{dx} - \tan x \cdot z = \sec x$$

$$\text{or } \frac{dz}{dx} + 3 \tan x \cdot z = -3 \sec x$$

$$P = 3 \tan x, \quad Q = -3 \sec x$$

$$I.F. = e^{\int 3 \tan x dx} = e^{3 \log \sec x} = e^{\log \sec^3 x} = \sec^3 x$$

Solution is given by

$$z \cdot \sec^3 x = -3 \int \sec^3 x \cdot \sec x dx$$

$$= -3 \int \sec^2 x \cdot \sec^2 x dx$$

$$= -3 \int (1 + \tan^2 x) \sec^2 x dx$$

$$= -3 \int (1 + t^2) dt$$

$$= -3 \left( t + \frac{t^3}{3} + \frac{C}{3} \right)$$

$$\left( \text{Put } t = \tan x \right. \\ \left. dt = \sec^2 x dx \right)$$

$$\text{or } \frac{1}{y^3} \sec^3 x = -3 \left( \tan x + \frac{\tan^3 x}{3} + \frac{C}{3} \right)$$

$$\text{or } \frac{\sec^3 x}{y^3} = -(\tan^3 x + 3 \tan x + C) \text{ is the soln.}$$

$$(4) \frac{dy}{dx} + \tan x \cdot \tan y = \cos x \cdot \sec y$$

Dividing by  $\sec y$

$$\frac{1}{\sec y} \frac{dy}{dx} + \frac{\tan x \cdot \tan y}{\sec y} = \cos x$$

$$\text{or } \cos y \frac{dy}{dx} + \tan x \cdot \sin y = \cos x \rightarrow (1)$$

$$\text{put } z = \sin y, \quad \frac{dz}{dx} = \cos y \frac{dy}{dx}$$

$$i. \frac{dz}{dx} + \tan x \cdot z = \cos x$$

$$P = \tan x, \quad Q = \cos x$$

$$IF = e^{\int \tan x dx} = e^{\log \sec x} = \sec x$$

$$\text{solution: } z (\sec x) = \int \sec x \cdot \cos x dx$$

$$i. z \sec x = \int 1 dx$$

$$\sin y \cdot \sec x = x + C$$

$$(5) \frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$$

Dividing by  $\cos^2 y$

$$\frac{1}{\cos^2 y} \frac{dy}{dx} + x \frac{\sin 2y}{\cos^2 y} = x^3$$

$$i. \sec^2 y \frac{dy}{dx} + x (2 \tan y) = x^3$$

$$\left( \because \frac{\sin 2y}{\cos^2 y} = \frac{2 \sin y \cos y}{\cos^2 y} = 2 \tan y \right)$$

$$\text{Put } z = \tan y, \quad \frac{dz}{dx} = \sec^2 y \frac{dy}{dx}$$

$$\therefore \text{ we get } \frac{dz}{dx} + 2xz = x^3 \quad \text{which is a linear eq}$$

$$P = 2x, \quad Q = x^3$$

$$IF = e^{\int 2x dx} = e^{x^2}$$

$$\text{solution } z \cdot e^{x^2} = \int e^{x^2} \cdot x^3 dx$$

$$= \int e^{x^2} \cdot x^2 \cdot x dx$$

$$= \int e^u \cdot u \frac{du}{2}$$

$$= \frac{1}{2} \int u e^u du$$

$$= \frac{1}{2} (u e^u - \int e^u du)$$

$$i. \tan y e^{x^2} = \frac{1}{2} (u e^u - e^u) + \frac{C}{2}$$

$$i. 2 \tan y e^{x^2} = e^u (u - 1) + C$$

$$= e^{x^2} (x^2 - 1) + C$$

$$i. 2 \tan y = (x^2 - 1) + C e^{-x^2}$$

Solve the following

$$(1) \frac{dy}{dx} + y = xy^3 \quad (2) \frac{dy}{dx} - y \tan x = y^2 \sec x$$

$$(3) \frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x} \log x \quad (4) \frac{dy}{dx} + xy = x^3 y^3$$

### Partial differentiation

If  $z$  is a function of  $x$  and  $y$  say  $f(x, y)$  it can be differentiated partially w.r. to  $x$  or partially w.r. to  $y$ .

Partial derivative of  $z$  w.r. to  $x$  is defined as

$$\frac{\partial z}{\partial x} = \lim_{\Delta x \rightarrow 0} \left[ \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} \right]$$

Here we treat  $y$  as a constant.

Partial derivative of  $z$  w.r. to  $y$  is defined as

$$\frac{\partial z}{\partial y} = \lim_{\Delta y \rightarrow 0} \left[ \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y} \right]$$

Here we treat  $x$  as a constant.

### Problems

Find  $\frac{\partial z}{\partial x}$  and  $\frac{\partial z}{\partial y}$  if

$$(i) z = x^3 + 3x^2y + xy^3$$

$$(ii) z = \tan^{-1}\left(\frac{y}{x}\right)$$

$$(iii) z = 2x \cos y + 3x^2y$$

$$(iv) z = x^3 - x^2 \sin y - y$$

$$(i) z = x^3 + 3x^2y + xy^3$$

$$\frac{\partial z}{\partial x} = 3x^2 + 6xy$$

$$\frac{\partial z}{\partial y} = 3x^2 + 3xy^2$$

$$(ii) z = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\frac{\partial z}{\partial x} = \frac{1}{\left(1 + \frac{y^2}{x^2}\right)} \frac{\partial}{\partial x} \left(\frac{y}{x}\right) = \frac{x^2}{x^2 + y^2} \left(-\frac{y}{x^2}\right) = \frac{-y}{x^2 + y^2}$$

$$\frac{\partial z}{\partial y} = \frac{1}{\left(1 + \frac{y^2}{x^2}\right)} \frac{\partial}{\partial y} \left(\frac{y}{x}\right) = \frac{x^2}{x^2 + y^2} \left(\frac{1}{x}\right) = \frac{x}{x^2 + y^2}$$



$$(iii) z = 2x \cos y + 3x^2 y$$

$$\frac{\partial z}{\partial x} = 2 \cos y + 6xy \quad \frac{\partial z}{\partial y} = -2x \sin y + 3x^2$$

Higher order partial derivatives

$$(i) \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial x} \right) \quad (ii) \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial y} \right)$$

$$(iii) \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial x} \right) \quad (iv) \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial y} \right)$$

If  $z = f(x, y)$  is a continuous function  
then  $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$

∴ order of differentiation is immaterial

Find first and second order partial derivatives of the following functions

$$(i) x^2 y$$

$$(ii) x^3 \sin y$$

$$(i) z = x^2 y$$

$$\frac{\partial z}{\partial x} = 2xy, \quad \frac{\partial z}{\partial y} = x^2, \quad \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} (2xy) = 2y$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} (x^2) = 0, \quad \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial y} (2xy) = 2x$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left( \frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial x} (x^2) = 2x$$

Here we notice that  $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$

$$(ii) z = x^3 \sin y$$

$$\frac{\partial z}{\partial x} = 3x^2 \sin y$$

$$\frac{\partial z}{\partial y} = x^3 \cos y$$

$$\frac{\partial^2 z}{\partial x^2} = 6x \sin y$$

$$\frac{\partial^2 z}{\partial y^2} = -x^3 \sin y$$

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left( \frac{\partial z}{\partial x} \right) = 3x^2 \cos y = \frac{\partial^2 z}{\partial x \partial y}$$

Differentials If  $z = f(x, y)$  then  $dx, dy$  and  $dz$  are called differentials in  $x, y$  and  $z$  respectively

Differential in  $z$  is given by

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

Example

$$z = x^2 y$$

$$\begin{aligned} dz &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \\ &= 2xy dx + x^2 dy \end{aligned}$$

1) Find differential in  $z$  if  $f(x, y) = \frac{x^3}{3} - xy^2$

$$\frac{\partial f}{\partial x} = x^2 - y^2, \quad \frac{\partial f}{\partial y} = 0 - 2xy = -2xy$$

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = (x^2 - y^2) dx - 2xy dy$$

Exact differential equation

A diff. eq of the form  $M(x, y) dx + N(x, y) dy = 0$  is said to be an exact diff. eq if there exists a function  $u(x, y)$  such that  $\frac{\partial u}{\partial x} = M(x, y)$  and  $\frac{\partial u}{\partial y} = N(x, y)$

In this case  $M dx + N dy = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$

$$\text{i.e. } M dx + N dy = du$$

This is called an exact differential

It's solution is given by  $u(x, y) = c$

Example consider the diff. eq

$$(1-x) dx - (1+y) dy = 0, \quad M = 1-x, \quad N = -1-y$$

$$\text{Let } u(x, y) = \left(x - \frac{x^2}{2}\right) - \left(y + \frac{y^2}{2}\right)$$

then  $\frac{\partial u}{\partial x} = 1-x = M$      $\frac{\partial u}{\partial y} = -(1+y) = N$

Given diff. eq is exact.

Solution is given by

$$u(x,y) = C \Rightarrow x - \frac{x^2}{2} - (y + \frac{y^2}{2}) = C$$

Condition for exactness

consider the diff. eq

$$M(x,y) dx + N(x,y) dy = 0 \text{ which is exact}$$

$\therefore$  There is a function  $u(x,y)$  such that

$$\frac{\partial u}{\partial x} = M \quad \text{and} \quad \frac{\partial u}{\partial y} = N$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial M}{\partial y} \quad \text{and} \quad \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial N}{\partial x}$$

$$\text{But } \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y} \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$  A diff. eq of the form  $M(x,y) dx + N(x,y) dy = 0$  is exact if  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Solution of an exact diff. eq.

$$\text{Solve } (x^2 - y^2) dx - 2xy dy = 0$$

$$M = x^2 - y^2, \quad N = -2xy$$

$$\frac{\partial M}{\partial y} = -2y \quad \frac{\partial N}{\partial x} = -2y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \therefore \text{eq is exact}$$

There is a function  $u(x,y)$  such that

$$\frac{\partial u}{\partial x} = M = x^2 - y^2 \quad \frac{\partial u}{\partial y} = N = -2xy \rightarrow (1)$$

$$u = \int M dx = \int (x^2 - y^2) dx = \frac{x^3}{3} - xy^2 + \phi(y) \rightarrow (2)$$

$$\therefore u(x,y) = \frac{x^3}{3} - xy^2 + \phi(y)$$

diff w.r. to y

$$\frac{\partial u}{\partial y} = -2xy + \phi'(y) \quad \text{ie } N = -2xy + \phi'(y)$$

But  $N = -2xy \quad \therefore \phi'(y) = 0 \quad \therefore \phi(y) = C$

sub in (1)  $u(x, y) = \frac{x^3}{3} - xy^2 + C$

$\therefore$  Soln of the given diff. eq is

$$u(x, y) = C_1$$

$$\text{ie } \frac{x^3}{3} - xy^2 + C = C_1 \Rightarrow \underline{\underline{\frac{x^3}{3} - xy^2 = C_1}}$$

Short-cut method

solution is given by

$$\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$$

Example  $(x^2 - y^2) dx - 2xy dy = 0$

$$M = x^2 - y^2 \quad N = -2xy$$

Soln:  $\int (x^2 - y^2) dx + \int 0 dy = C$

$$\text{ie } \underline{\underline{\frac{x^3}{3} - xy^2 = C}}$$

Solve  $(3x^2 + 4xy) dx + (2x^2 + 2y) dy = 0$

$$M = 3x^2 + 4xy, \quad N = 2x^2 + 2y$$

$$\frac{\partial M}{\partial y} = 4x, \quad \frac{\partial N}{\partial x} = 4x$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \therefore \text{Eq is exact}$$

Soln:  $\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$

$$\text{ie } \int (3x^2 + 4xy) dx + \int 2y dy = C$$

$$\text{ie } \frac{3x^3}{3} + 4 \frac{x^2 y}{2} + y^2 = C$$

$$\text{ie } x^3 + 2x^2 y + y^2 = C$$



Differentials

If  $z = f(x, y)$  then  $dx, dy$  and  $dz$  are called differentials in  $x, y$  and  $z$  respectively

Differential in  $z$  is given by

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

Example

$$z = x^2 y$$

$$\begin{aligned} dz &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \\ &= 2xy dx + x^2 dy \end{aligned}$$

1) Find differential in  $z$  if  $f(x, y) = \frac{x^3}{3} - xy^2$

$$\frac{\partial f}{\partial x} = x^2 - y^2, \quad \frac{\partial f}{\partial y} = 0 - 2xy = -2xy$$

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = (x^2 - y^2) dx - 2xy dy$$

Exact differential equation

A diff. eq of the form  $M(x, y) dx + N(x, y) dy = 0$  is said to be an exact diff. eq if there exists a function  $u(x, y)$  such that  $\frac{\partial u}{\partial x} = M(x, y)$  and  $\frac{\partial u}{\partial y} = N(x, y)$

$$\text{In this case } M dx + N dy = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$$\text{or } M dx + N dy = du$$

This is called an exact differential

Its solution is given by  $u(x, y) = c$

Example consider the diff. eq

$$(1-x) dx - (1+y) dy = 0, \quad M = 1-x, \quad N = -1-y$$

$$\text{Let } u(x, y) = \left(x - \frac{x^2}{2}\right) - \left(y + \frac{y^2}{2}\right)$$

then  $\frac{\partial u}{\partial x} = 1-x = M$      $\frac{\partial u}{\partial y} = -(1+y) = N$

Given diff. eq is exact.

Solution is given by

$$u(x, y) = C \Rightarrow x - \frac{x^2}{2} - (y + \frac{y^2}{2}) = C$$

Condition for exactness

consider the diff. eq

$$M(x, y) dx + N(x, y) dy = 0 \text{ which is exact}$$

$\therefore$  There is a function  $u(x, y)$  such that

$$\frac{\partial u}{\partial x} = M \quad \text{and} \quad \frac{\partial u}{\partial y} = N$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial M}{\partial y} \quad \text{and} \quad \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial N}{\partial x}$$

$$\text{But } \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y} \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$  A diff. eq of the form  $M(x, y) dx + N(x, y) dy = 0$  is exact if  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Solution of an exact diff. eq.

$$\text{Solve } (x^2 - y^2) dx - 2xy dy = 0$$

$$M = x^2 - y^2, \quad N = -2xy$$

$$\frac{\partial M}{\partial y} = -2y \quad \frac{\partial N}{\partial x} = -2y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \therefore \text{eq is exact}$$

There is a function  $u(x, y)$  such that

$$\frac{\partial u}{\partial x} = M = x^2 - y^2 \quad \frac{\partial u}{\partial y} = N = -2xy \rightarrow (1)$$

$$u = \int M dx = \int (x^2 - y^2) dx = \frac{x^3}{3} - xy^2 + \phi(y) \rightarrow (2)$$

$$\therefore u(x, y) = \frac{x^3}{3} - xy^2 + \phi(y)$$

diff w.r. to  $y$

$$\frac{\partial u}{\partial y} = -2xy + \phi'(y) \quad \text{ie } N = -2xy + \phi'(y)$$

But  $N = -2xy \quad \therefore \phi'(y) = 0 \quad \therefore \phi(y) = C$

Sub in (1)  $u(x, y) = \frac{x^3}{3} - xy^2 + C$

$\therefore$  Soln of the given diff. eq is

$$u(x, y) = C_1$$

$$\text{ie } \frac{x^3}{3} - xy^2 + C = C_1 \Rightarrow \underline{\underline{\frac{x^3}{3} - xy^2 = C_1}}$$

Short-cut method

Solution is given by

$$\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$$

Example (1)  $(x^2 - y^2) dx - 2xy dy = 0$

$$M = x^2 - y^2 \quad N = -2xy$$

Soln:  $\int (x^2 - y^2) dx + \int 0 dy = C$

$$\text{ie } \underline{\underline{\frac{x^3}{3} - xy^2 = C}}$$

(2) Solve  $(3x^2 + 4xy) dx + (2x^2 + 2y) dy = 0$

$$M = 3x^2 + 4xy, \quad N = 2x^2 + 2y$$

$$\frac{\partial M}{\partial y} = 4x, \quad \frac{\partial N}{\partial x} = 4x$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \therefore \text{Eq is exact}$$

Soln:  $\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$

$$\text{ie } \int (3x^2 + 4xy) dx + \int 2y dy = C$$

$$\text{ie } \frac{3x^3}{3} + 4 \frac{x^2 y}{2} + y^2 = C$$

$$\text{ie } x^3 + 2x^2 y + y^2 = C$$

Solve the following diff eqs

$$(1) (3x^2 + 4xy) dx + (2x^2 + 2y) dy = 0$$

$$(2) (2x \cos y + 3x^2 y) dx + (x^3 - x^2 \sin y - y) dy = 0$$

$$(3) (5x^4 + 3x^2 y^2 - 2xy^3) dx + (2x^3 y - 3x^2 y^2 - 5y^4) dy = 0$$

$$(4) (2xy + y - \tan y) dx + (x^2 - x \tan^2 y + \sec^2 y + 2) dy = 0$$

$$(2) (2x \cos y + 3x^2 y) dx + (x^3 - x^2 \sin y - y) dy = 0$$

$$M = 2x \cos y + 3x^2 y, \quad N = x^3 - x^2 \sin y - y$$

$$\frac{\partial M}{\partial y} = -2x \sin y + 3x^2, \quad \frac{\partial N}{\partial x} = 3x^2 - 2x \sin y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ Hence eq. is exact}$$

Solution is given by

$$\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$$

$$\therefore \int (2x \cos y + 3x^2 y) dx + \int (-y) dy = C$$

$$\therefore x^2 \cos y + x^3 y - \frac{y^2}{2} = C$$

$$(3) (5x^4 + 3x^2 y^2 - 2xy^3) dx + (2x^3 y - 3x^2 y^2 - 5y^4) dy = 0$$

$$M = 5x^4 + 3x^2 y^2 - 2xy^3, \quad N = 2x^3 y - 3x^2 y^2 - 5y^4$$

$$\frac{\partial M}{\partial y} = 6x^2 y - 6xy^2, \quad \frac{\partial N}{\partial x} = 6x^2 y - 6xy^2$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ Hence eq. is exact}$$

Solution is given by

$$\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$$

$$\therefore \int (5x^4 + 3x^2 y^2 - 2xy^3) dx + \int (-5y^4) dy = C$$

$$\therefore x^5 + x^3 y^2 - x^2 y^3 - y^5 = C$$



$$(4) (2xy + y - \tan y) dx + (x^2 - x \tan^2 y + \sec^2 y + 2) dy = 0$$

$$M = 2xy + y - \tan y, \quad N = x^2 - x \tan^2 y + \sec^2 y + 2$$

$$\frac{\partial M}{\partial y} = 2x + 1 - \sec^2 y$$

$$= 2x + 1 - (1 + \tan^2 y)$$

$$= \underline{2x - \tan^2 y}$$

$$\frac{\partial N}{\partial x} = \underline{2x - \tan^2 y}$$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ . Hence eq is exact

Soln is given by

$$\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$$

$$\text{i.e. } \int (2xy + y - \tan y) dx + \int (\sec^2 y + 2) dy = C$$

$$\text{i.e. } x^2 y + xy - x \tan y + \tan y + 2y = C$$

$$\text{i.e. } x^2 y + xy + (1-x) \tan y + 2y = C$$

(5) Under what condition the diff. eq

$$(ax + by) dx + (lx + my) dy = 0 \text{ is exact?}$$

Solve the resulting diff. eq.

$$M = ax + by, \quad N = lx + my$$

$$\frac{\partial M}{\partial y} = b, \quad \frac{\partial N}{\partial x} = l$$

The eq is exact if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow b = l \text{ is the required condition. Resulting diff eq is}$$

$$(ax + by) dx + (bx + my) dy = 0 \text{ which is exact}$$

Solution is given by

$$\int (ax + by) dx + \int my dy = 0$$

$$\text{i.e. } a \frac{x^2}{2} + bxy + \frac{my^2}{2} = \frac{C}{2} \Rightarrow ax^2 + 2bxy + my^2 = C$$

(6) Test the exactness and solve the initial value problem

$$ye^x dx + (2y + e^x) dy = 0, \quad y(0) = -1$$

$$M = ye^x$$

$$N = 2y + e^x$$

$$\frac{\partial M}{\partial y} = e^x$$

$$\frac{\partial N}{\partial x} = e^x$$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$  Hence the eq is exact

Solution is given by

$$\int M dx + \int (\text{terms of } N \text{ not involving } x) dy = C$$

$$\therefore \int ye^x dx + \int 2y dy = C$$

$$\therefore ye^x + y^2 = C \rightarrow (1) \text{ is the general solution}$$

Given that  $y(0) = -1$ . This means that when  $x=0$ ,  $y=-1$

$\therefore$  From eq (1) we get,

$$-1 \cdot e^0 + 1 = C \Rightarrow C = -1 + 1 = 0$$

Required solution is given by

$$\underline{ye^x + y^2 = 0}$$

Solve the following diff. eqs.

$$(1) (y \cos x + x) dx + (\sin x + y) dy = 0$$

$$(2) (3x^2 + 6xy^2) dx + (6x^2y + 4y^3) dy = 0$$

$$(3) e^x dx + (xe^y + xy) dy = 0$$

$$(4) \frac{y}{x^2} dx + (y - \frac{1}{x}) dy = 0$$

### Simple applications

slope at any point  $(x, y)$  of a curve is given by  $1 + \frac{y}{x}$ . If the curve passes through  $(1, 1)$  find its equation

Given that  $\frac{dy}{dx} = 1 + \frac{y}{x} \rightarrow (1)$

This is a homogeneous diff. eq

Put  $y = vx$   $\therefore \frac{dy}{dx} = v + x \frac{dv}{dx}$

sub in (1)  $v + x \frac{dv}{dx} = 1 + v$

$\therefore x \frac{dv}{dx} = 1 \Rightarrow dv = \frac{dx}{x}$

Integrating,  $v = \log x + c$

$\therefore \frac{y}{x} = \log x + c \rightarrow (2)$

This curve passes through  $(1, 1)$

$\therefore 1 = \log 1 + c \Rightarrow c = 1 \quad (\because \log 1 = 0)$

sub in (2),  $\frac{y}{x} = \log x + 1$

$\therefore y = x(\log x + 1)$  is the required

Solution

(2) If the temperature of a cake is  $300^\circ\text{F}$  when it is taken from oven and  $200^\circ\text{F}$  ten minutes later. When will it be practically equal to the room temperature of  $60^\circ\text{F}$ ?

Solution According to Newton's law of cooling, rate of change of temperature  $T$  is proportional to difference between  $T$  and the temp. of the surroundings  $60^\circ\text{F}$

$\therefore \frac{dT}{dt} = k(T - 60)$  where  $k$  is a constant

ie  $\frac{dT}{dt} - KT = -60K$  This is a linear diff. eq.

$$I.F = e^{\int -K dt} = e^{-Kt}$$

general soln:  $T(e^{-Kt}) = \int e^{-Kt} \cdot (-60K) dt$

$$T \cdot e^{-Kt} = -60K \frac{e^{-Kt}}{-K} + C$$

$$ie \quad T e^{-Kt} = 60 e^{-Kt} + C$$

$$\therefore T = 60 + C e^{Kt} \rightarrow (1)$$

Initially  $T(0) = 300$  ie  $T = 300$  when  $t = 0$

$$\therefore 300 = C e^0 + 60$$

$$ie \quad C = 240$$

Sub in eq (1)  $T = 60 + 240 e^{Kt} \rightarrow (2)$

Also  $T(10) = 200$  ie  $T = 200$  when  $t = 10$

$$\therefore 200 = 60 + 240 e^{10K}$$

$$\Rightarrow 140 = 240 e^{10K} \Rightarrow \frac{140}{240} = e^{10K}$$

$$\Rightarrow \frac{7}{12} = e^{10K} \Rightarrow 10K = \log\left(\frac{7}{12}\right)$$

$$\Rightarrow K = \frac{\log\left(\frac{7}{12}\right)}{10} = -0.0539$$

Sub in eq (2)  $T = 60 + 240 e^{-(0.539)t}$   
This is the temp. of the cake at any time t.



## Higher order linear differential equations

The general form of  $n^{\text{th}}$  order linear diff. eq is given by

$$P_0(x) \frac{d^n y}{dx^n} + P_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + P_{n-1}(x) \frac{dy}{dx} + P_n(x) y = \phi(x)$$

where  $P_0(x), P_1(x), \dots, P_n(x)$  and  $\phi(x)$  are functions of  $x$

### Linear diff. eq. with constant coefficients

A diff. eq. of the form

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = \phi(x) \rightarrow (I)$$

where  $a_0, a_1, a_2, \dots, a_n$  are constants, is the most general form of a linear diff. eq with const. coeffs.

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = \phi(x) \rightarrow (II) \text{ is the general}$$

form of a second order linear diff. eq with const coeffs.

Associated diff. eq. of eq (II) is given by

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0 \rightarrow (III)$$

It is also called homogeneous part of eq (II)

### Solution of second order linear homogeneous equation

Consider the diff eq.

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0$$

Dividing by  $a_0$  we get

$$\frac{d^2 y}{dx^2} + \frac{a_1}{a_0} \frac{dy}{dx} + \frac{a_2}{a_0} y = 0$$

$$\text{or } \frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Q y = 0 \rightarrow (I) \quad (P = \frac{a_1}{a_0}, Q = \frac{a_2}{a_0})$$

since this is a diff eq with const coeff  
we assume a trial solution  $y = e^{mx}$

$$\frac{dy}{dx} = m e^{mx} \quad \frac{d^2y}{dx^2} = m^2 e^{mx}$$

sub in (1)

$$m^2 e^{mx} + p m e^{mx} + q e^{mx} = 0$$

$$e^{mx} (m^2 + pm + q) = 0$$

This is true only if

$$m^2 + pm + q = 0 \rightarrow (2) \quad (\because e^{mx} \neq 0)$$

This is quadratic eq in  $m$ . It has two roots.  $\Delta = p^2 - 4q$  is the discriminant

Case (I)  $p^2 - 4q > 0$ . eq (2) has two distinct real roots say  $m_1$  and  $m_2$ .  $\therefore e^{m_1 x}$  and  $e^{m_2 x}$  are independent solutions

$\therefore y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$  is the gen. soln.

Case (II)  $p^2 - 4q = 0$  eq (2) has two equal roots. i.e.  $m_2 = m_1$ . In this case we get two independent solutions  $e^{m_1 x}$  and  $x e^{m_1 x}$

$\therefore$  General solution of (2) is given by

$$y = c_1 e^{m_1 x} + c_2 x e^{m_1 x} = (c_1 + c_2 x) e^{m_1 x}$$

Case (III)  $p^2 - 4q < 0$ . eq (2) has two complex roots say  $\alpha \pm i\beta$ . The gen. soln is given

General solution is given by

$$y = c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x}$$

$$= c_1 e^{\alpha x} \cdot e^{i\beta x} + c_2 e^{\alpha x} \cdot e^{-i\beta x}$$

$$= e^{\alpha x} [c_1 e^{i\beta x} + c_2 e^{-i\beta x}]$$

$$= e^{\alpha x} [c_1 (\cos \beta x + i \sin \beta x) + c_2 (\cos \beta x - i \sin \beta x)]$$

$$\bar{w} \quad y = e^{\alpha x} [(c_1 + c_2) \cos \beta x + i(c_1 - c_2) \sin \beta x]$$

$$\bar{w} \quad y = e^{\alpha x} [C_1 \cos \beta x + C_2 \sin \beta x]$$

$$\text{where } C_1 = c_1 + c_2 \quad C_2 = i(c_1 - c_2)$$

Solve the following diff. eqs

$$(1) \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = 0$$

$$(2) \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0$$

$$(3) \frac{d^2 y}{dx^2} + 4y = 0$$

$$(4) \frac{d^2 y}{dx^2} + 10 \frac{dy}{dx} + 29y = 0$$

$$(1) \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = 0$$

$$\text{Let } D \equiv \frac{d}{dx} \text{ and } D^2 \equiv \frac{d^2}{dx^2}$$

$$\therefore \text{ we get } D^2 y + 5Dy + 6y = 0 \Rightarrow (D^2 + 5D + 6)y = 0$$

$$\bar{w} \quad D^2 + 5D + 6 = 0 \text{ is the Auxillary eq.}$$

$$\bar{w} \quad (D+2)(D+3) = 0 \Rightarrow D = -2, -3$$

General solution is given by

$$y = C_1 e^{-2x} + C_2 e^{-3x}$$

$$(2) \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0 \quad \bar{w} \quad D^2 y - 2Dy + y = 0$$

$$\Rightarrow (D^2 - 2D + 1)y = 0$$

Auxillary eq is given by

$$D^2 - 2D + 1 = 0 \Rightarrow (D-1)^2 = 0 \quad D = 1, 1$$

$$\text{General soln, } y = (C_1 + C_2 x) \cdot e^x$$

$$(3) \frac{d^2 y}{dx^2} + 4y = 0 \Rightarrow (D^2 + 4)y = 0$$

$$\text{Auxillary eq is } D^2 + 4 = 0 \Rightarrow D^2 = -4 \Rightarrow D = \pm 2i$$

$$\bar{w} \quad D = 0 \pm 2i \quad (\alpha = 0, \beta = 2)$$

General solution

$$\begin{aligned} y &= e^{\alpha x} [C_1 \cos \beta x + C_2 \sin \beta x] \\ &= e^{0x} [C_1 \cos 2x + C_2 \sin 2x] \\ &= C_1 \cos 2x + C_2 \sin 2x \end{aligned}$$

$$(4) \frac{d^2 y}{dx^2} + 10 \frac{dy}{dx} + 29y = 0 \quad \text{u} \quad (\beta^2 + 10\beta + 29)y = 0$$

Auxiliary eq is  $\beta^2 + 10\beta + 29 = 0$

$$D = \frac{-10 \pm \sqrt{100 - 4 \times 29}}{2} = \frac{-10 \pm \sqrt{-16}}{2}$$

$$= \frac{-10 \pm 4i}{2} = -5 \pm 2i$$

( $\alpha = -5$ ,  $\beta = 2$ )

General soln:

$$y = e^{\alpha x} [c_1 \cos \beta x + c_2 \sin \beta x]$$

$$= e^{-5x} [c_1 \cos 2x + c_2 \sin 2x]$$

Solve the following diff. eqs

$$(1) \frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} + 100 \frac{dy}{dx} - 100y = 0$$

$$(2) \frac{d^3 y}{dx^3} - 4 \frac{d^2 y}{dx^2} + \frac{dy}{dx} + 6y = 0$$

$$(3) (D^2 + D + 1)y = 0$$

$$(4) \frac{d^4 x}{dt^4} + 4x = 0$$

Answers

$$(1) (D^3 - D^2 + 100D - 100)y = 0$$

$$\text{A.E. is } D^3 - D^2 + 100D - 100 = 0$$

$$\Rightarrow D^2(D-1) + 100(D-1) = 0$$

$$\Rightarrow (D^2 + 100)(D-1) = 0$$

$$D^2 + 100 = 0 \Rightarrow D^2 = -100 \Rightarrow D = \pm 10i$$

$$\text{Also } D-1 = 0 \Rightarrow D = 1$$

Hence the roots are  $D = 1, \pm 10i$

General solution is given by

$$y = C_1 e^x + e^{0x} [C_2 \cos 10x + C_3 \sin 10x] \quad (\alpha = 0, \beta = 10)$$

$$\text{u } y = C_1 e^x + C_2 \cos 10x + C_3 \sin 10x$$



$$(2) \frac{d^3 y}{dx^3} - 4 \frac{d^2 y}{dx^2} + \frac{dy}{dx} + 6y = 0$$

$$\text{i.e. } (D^3 - 4D^2 + D + 6)y = 0$$

Auxiliary eq. is given by

$$D^3 - 4D^2 + D + 6 = 0$$

$D = -1$  is a root

$\therefore D+1$  is a factor

Dividing  $D^3 - 4D^2 + D + 6$  by  $D+1$

we get  $(D^2 - 5D + 6)$

$$\begin{aligned} \therefore D^3 - 4D^2 + D + 6 &= (D+1)(D^2 - 5D + 6) \\ &= (D+1)(D-2)(D-3) \end{aligned}$$

$$\therefore D = 1, 2, 3$$

Hence gen. soln is given by

$$y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x}$$

$$(3) (D^2 + D + 1)y = 0$$

$$\text{A.E. is } D^2 + D + 1 = 0$$

$$D = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{3}i}{2}$$

General solution is given by  $(\alpha = -\frac{1}{2}, \beta = \frac{\sqrt{3}}{2})$

$$y = e^{\alpha x} [c_1 \cos \beta x + c_2 \sin \beta x]$$

$$= e^{-\frac{x}{2}} \left[ c_1 \cos \frac{\sqrt{3}x}{2} + c_2 \sin \frac{\sqrt{3}x}{2} \right]$$

$$(4) \frac{d^4 x}{dt^4} + 4x = 0 \Rightarrow (D^4 + 4)x = 0$$

$$\text{A.E. is given by } D^4 + 4 = 0$$

$$\text{i.e. } (D^4 + 4D^2 + 4) - 4D^2 = 0$$

$$\text{i.e. } (D^2 + 2)^2 - (2D)^2 = 0$$

$$\begin{array}{r} D^2 - 5D + 6 \\ D+1 \overline{) D^3 - 4D^2 + D + 6} \\ \underline{D^3 + D^2} \phantom{+ 6} \\ -5D^2 + D \phantom{+ 6} \\ \underline{-5D^2 - 5D} \phantom{+ 6} \\ 6D + 6 \\ \underline{6D + 6} \\ 0 \end{array}$$

$$i \quad (D^2 + 2D + 2)(D^2 + 2 - 2D) = 0$$

$$\Rightarrow D^2 + 2D + 2 = 0,$$

$$i \quad D = \frac{-2 \pm \sqrt{4-8}}{2}$$

$$= \frac{-2 \pm \sqrt{-4}}{2}$$

$$= \frac{-2 \pm 2i}{2}$$

$$= -1 \pm i$$

$$(\alpha = -1, \beta = 1)$$

$$D^2 - 2D + 2 = 0$$

$$D = \frac{2 \pm \sqrt{4-8}}{2}$$

$$= \frac{2 \pm 2i}{2}$$

$$= 1 \pm i$$

$$(\alpha = 1, \beta = 1)$$

General solution:

$$y = e^{-t} [C_1 \cos t + C_2 \sin t] + e^t [C_3 \cos t + C_4 \sin t]$$

Solve the following diff. eqs.

$$(1) \frac{d^4 y}{dx^4} - y = 0$$

$$(2) (D^2 - 6D + 9)y = 0$$

$$(3) \frac{d^3 y}{dx^3} - 13 \frac{dy}{dx} + 12y = 0$$

$$(4) (D^3 + 1)y = 0$$

$$(5) \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 5y = 0$$

Solution of non-homogeneous diff. eq.

Consider the diff. eq.

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = \phi(x)$$

$$i \quad (a_0 D^2 + a_1 D + a_2)y = \phi(x) \rightarrow (1)$$

$$ii \quad f(D)y = \phi(x)$$

$$\text{where } f(D) = a_0 D^2 + a_1 D + a_2$$

Homogeneous part of this eq is given by

$$(a_0 D^2 + a_1 D + a_2)y \rightarrow (2)$$

$$i \quad f(D)y = 0$$

General solution of eq (2) is called complementary function (CF) and any solution of (2) is called a particular integral (PI)

### Theorem

Consider the diff. eq

$$(a_0 D^2 + a_1 D + a_2)y = \phi(x) \rightarrow (1)$$

It's homogeneous part is given by

$$(a_0 D^2 + a_1 D + a_2)y = 0 \rightarrow (2)$$

General solution of eq (1) is given by

$$y = CF + PI$$

Proof Let  $y_g$  be the general soln of eq (2)

$$\text{i.e. } (a_0 D^2 + a_1 D + a_2)y_g = 0 \rightarrow (3)$$

and  $y_p$  be a particular soln of eq (1)

$$\text{i.e. } (a_0 D^2 + a_1 D + a_2)y_p = \phi(x) \rightarrow (4)$$

$$\text{Let } y = y_g + y_p \rightarrow (5)$$

$$\begin{aligned} (a_0 D^2 + a_1 D + a_2)y &= (a_0 D^2 + a_1 D + a_2)(y_g + y_p) \\ &= (a_0 D^2 + a_1 D + a_2)y_g + (a_0 D^2 + a_1 D + a_2)y_p \\ &= 0 + \phi(x) \quad (\text{using (3) and (4)}) \\ &= \phi(x) \end{aligned}$$

This shows that  $y = y_g + y_p$  satisfies eq (1)

$\therefore$  It is a solution of eq (1).

Since  $y_g$  is the gen. soln of eq (2)

it contains two arbitrary constants

eq (3) gives the gen. soln of (1).

$$\text{i.e. } y = y_g + y_p = CF + PI \text{ is the gen. soln.}$$

## Special methods of finding Particular integral

consider the diff. eq

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = \phi(x)$$

$$\text{i.e. } f(D)y = \phi(x) \quad \text{where } f(D) = a_0 D^2 + a_1 D + a_2 \quad \text{--- (1)}$$

It's gen. soln is given by

$$y = CF + PI$$

case (I)  $\phi(x) = e^{ax}$

$$D(e^{ax}) = a e^{ax}, \quad D^2(e^{ax}) = a^2 e^{ax}$$

$$\begin{aligned} \therefore f(D)(e^{ax}) &= (a_0 D^2 + a_1 D + a_2) e^{ax} \\ &= a_0 a^2 e^{ax} + a_1 a e^{ax} + a_2 e^{ax} \\ &= (a_0 a^2 + a_1 a + a_2) e^{ax} \\ &= f(a) e^{ax} \end{aligned}$$

Operating both sides by  $f(D)^{-1} = \frac{1}{f(D)}$ , we get

$$\frac{1}{f(D)} \cdot f(D) e^{ax} = \frac{1}{f(D)} f(a) e^{ax}$$

$$\text{i.e. } e^{ax} = f(a) \cdot \frac{1}{f(D)} e^{ax}$$

$$\therefore \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax}$$

provided  $f(a) \neq 0$ .

$$\text{i.e. } PI = \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax} \quad \text{if } f(a) \neq 0$$

Note  $f(a) = 0 \Rightarrow a$  is a root of the A.E.  
 $f(D) = 0$

If 'a' is a root of multiplicity  $r$  then

$$PI = \frac{x^r}{f^{(r)}(a)} e^{ax} \quad \text{where } f^{(r)}(a) = f^{(r)}(D) \text{ at } D=a$$

$$\text{when } r=1, \quad PI = \frac{x}{f'(a)} e^{ax}$$

$$\text{when } r=2 \quad PI = \frac{x^2}{f''(a)} e^{ax} \dots$$



Examples

(1) Solve  $\frac{d^2 y}{dx^2} - 13 \frac{dy}{dx} + 12y = e^{-2x}$

Associated eq is given by

$$\frac{d^2 y}{dx^2} - 13 \frac{dy}{dx} + 12y = 0 \Rightarrow (D^2 - 13D + 12)y = 0$$

A.E. is given by  $D^2 - 13D + 12 = 0$

$$\therefore (D-1)(D-12) = 0 \Rightarrow D = 1, 12$$

$$\therefore CF = C_1 e^x + C_2 e^{12x}$$

$$PI = \frac{1}{f(D)} e^{-2x} = \frac{1}{D^2 - 13D + 12} e^{-2x} \quad (a = -2)$$

$$= \frac{1}{(-2)^2 - 13(-2) + 12} e^{-2x} \quad \left( \begin{array}{l} \text{Replacing} \\ D \text{ by } -2 \end{array} \right)$$

$$= \frac{1}{4 + 26 + 12} e^{-2x}$$

$$= \frac{1}{42} e^{-2x}$$

Gen. soln. is given by

$$y = CF + PI = C_1 e^x + C_2 e^{12x} + \frac{1}{42} e^{-2x}$$

(2) Solve  $\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} - 10y = e^{2x}$

$$\therefore (D^2 + 3D - 10)y = e^{2x}$$

A.E. is  $D^2 + 3D - 10 = 0$

$$\therefore (D-2)(D+5) = 0 \Rightarrow D = 2, -5$$

$$CF = C_1 e^{2x} + C_2 e^{-5x}$$

$$PI = \frac{1}{f(D)} e^{2x} = \frac{1}{D^2 + 3D - 10} e^{2x} \quad \left( \begin{array}{l} a=2 \text{ which is} \\ \text{single root of A.E.} \end{array} \right)$$

$$= \frac{x}{f'(a)} e^{2x}$$

$$= \frac{x}{7} e^{2x}$$

$$\left\{ \begin{array}{l} f(D) = D^2 + 3D - 10 \\ f'(D) = 2D + 3 \\ f'(a) = 2(2) + 3 = 7 \end{array} \right.$$

gen. soln.  $y = CF + PI = C_1 e^{2x} + C_2 e^{-5x} + \frac{x}{7} e^{2x}$

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$$3) \frac{d^2 y}{dx^2} - 12 \frac{dy}{dx} + 36y = e^{6x}$$

$$\text{or } (D^2 - 12D + 36)y = e^{6x}$$

$$\text{A.E. is } D^2 - 12D + 36 = 0$$

$$\text{or } (D-6)^2 = 0 \Rightarrow D = 6, 6$$

$$\text{CF} = (C_1 + C_2 x) e^{6x}$$

$$\text{PI} = \frac{1}{D^2 - 12D + 36} e^{6x}$$

$$= \frac{x^2}{f''(a)} e^{6x}$$

$$= \frac{x^2}{2} e^{6x}$$

( $a=6$  which is a double root of the A.E.)

$$\left| \begin{array}{l} f(D) = D^2 - 12D + 36 \\ f'(D) = 2D - 12 \\ f''(D) = 2 \therefore f''(6) = 2 \end{array} \right|$$

gen. soln is  $y = \text{CF} + \text{PI}$

$$= (C_1 + C_2 x) e^{6x} + \frac{x^2}{2} e^{6x}$$

Solve (1)  $\frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 5y = 2e^x + 10e^{-5x}$

(2)  $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 4y = 4e^x - 18e^{-x}$

(3)  $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^x + 4e^{3x}$

Answers

(1)  $(D^2 + 6D + 5)y = 2e^x + 10e^{-5x}$

$$\text{A.E. is } D^2 + 6D + 5 = 0$$

$$\text{or } (D+1)(D+5) = 0 \Rightarrow D = -1, -5$$

$$\text{CF} = C_1 e^{-x} + C_2 e^{-5x}$$

$$\text{PI} = \frac{1}{D^2 + 6D + 5} (2e^x + 10e^{-5x})$$

$$= \frac{1}{D^2 + 6D + 5} (2e^x) + \frac{1}{D^2 + 6D + 5} (10e^{-5x})$$

$$= \text{PI}_1 + \text{PI}_2$$

$$\text{PI}_1 = \frac{1}{D^2 + 6D + 5} (2e^x) = \frac{1}{1^2 + 6(1) + 5} (2e^x) \quad (a=1 \text{ which is not a root of A.E.})$$

$$PI_1 = \frac{1}{12} (2e^x) = \frac{1}{6} e^x$$

$$PI_2 = \frac{1}{D^2 + bD + 5} (10e^{-5x})$$

$$= \frac{x}{f'(a)} (10e^{-5x})$$

$$= \frac{x}{-4} (10e^{-5x})$$

$$= -\frac{5x}{2} e^{-5x}$$

( $a = -5$  which is a single root of the A.E.)

$$f(D) = D^2 + bD + 5$$

$$f'(D) = 2D + b$$

$$f'(a) = 2(-5) + b = -4$$

gen. soln  $y = CF + PI$

$$y = CF + PI_1 + PI_2$$

$$\therefore y = c_1 e^{-x} + c_2 e^{-5x} + \frac{1}{6} e^x - \frac{5x}{2} e^{-5x}$$

$$(2) (D^2 - 3D + 4)y = 4e^x - 18e^{-x}$$

A.E. is  $D^2 - 3D + 4 = 0$

$$D = \frac{3 \pm \sqrt{9 - 16}}{2} = \frac{3 \pm i\sqrt{7}}{2}$$

$$(\alpha = \frac{3}{2}, \beta = \frac{\sqrt{7}}{2})$$

$$CF = e^{\alpha x} [c_1 \cos \beta x + c_2 \sin \beta x]$$

$$= e^{\frac{3x}{2}} [c_1 \cos \frac{\sqrt{7}}{2} x + c_2 \sin \frac{\sqrt{7}}{2} x]$$

$$PI = \frac{1}{D^2 - 3D + 4} (4e^x - 18e^{-x})$$

$$= \frac{1}{D^2 - 3D + 4} (4e^x) + \frac{1}{D^2 - 3D + 4} (-18e^{-x})$$

$$= \frac{1}{D^2 - 3D + 4} (4e^x) + \frac{1}{(-1)^2 - 3(-1) + 4} (-18e^{-x})$$

$$= \frac{1}{2} (4e^x) + \frac{1}{8} (-18e^{-x})$$

$$= 2e^x - \frac{9}{4} e^{-x}$$

gen. soln  $y = CF + PI$

$$= e^{\frac{3x}{2}} [c_1 \cos \frac{\sqrt{7}}{2} x + c_2 \sin \frac{\sqrt{7}}{2} x] + 2e^x - \frac{9}{4} e^{-x}$$

$$(3) (D^2 - 3D + 2)y = e^x + 4e^{3x}$$

$$\text{A.E. is } D^2 - 3D + 2 = 0 \Rightarrow (D-1)(D-2) = 0$$

$$\therefore D = 1, 2$$

$$CF = c_1 e^x + c_2 e^{2x}$$

$$PI = \frac{1}{D^2 - 3D + 2} (e^x + 4e^{3x})$$

$$= \frac{1}{D^2 - 3D + 2} (e^x) + \frac{1}{D^2 - 3D + 2} (4e^{3x})$$

$$= PI_1 + PI_2$$

$$PI_1 = \frac{1}{D^2 - 3D + 2} (e^x)$$

(Here  $a=1$  which is a single root of the A.E)

$$= \frac{x}{f'(a)} e^{ax}$$

$$= \frac{x}{-1} e^x = -xe^x$$

$$\left( \begin{array}{l} f(D) = D^2 - 3D + 2 \\ f'(D) = 2D - 3 \\ f'(a) = 2(1) - 3 = -1 \end{array} \right)$$

$$PI_2 = \frac{1}{D^2 - 3D + 2} (4e^{3x})$$

(Here  $a=3$  which is not a root of the A.E)

$$= \frac{1}{3^2 - 3(3) + 2} (4e^{3x}) = \frac{1}{2} (4e^{3x}) = 2e^{3x}$$

$$\text{Gen. Soln } y = CF + PI = c_1 e^x + c_2 e^{2x} - xe^x + 2e^{3x}$$

Solve the following diff. eqs

$$(1) (D^2 - 5D + 6)y = e^{5x}$$

$$(2) (D^2 + 5D + 7)y = e^{-x}$$

$$(3) (D^2 + 4D + 4)y = e^{-3x}$$

$$(4) \frac{d^2 y}{dx^2} - 9y = 5e^{4x} + 2e^{3x}$$

$$(5) (D^2 - 6D + 8)y = (e^{2x} - 1)^2$$



$$(D^2 + 2D + 2)(D^2 + 2 - 2D) = 0$$

$$\Rightarrow D^2 + 2D + 2 = 0,$$

$$\therefore D = \frac{-2 \pm \sqrt{4 - 8}}{2}$$

$$= \frac{-2 \pm \sqrt{-4}}{2}$$

$$= \frac{-2 \pm 2i}{2}$$

$$= -1 \pm i$$

$$(\alpha = -1, \beta = 1)$$

$$D^2 - 2D + 2 = 0$$

$$D = \frac{2 \pm \sqrt{4 - 8}}{2}$$

$$= \frac{2 \pm 2i}{2}$$

$$= 1 \pm i$$

$$(\alpha = 1, \beta = 1)$$

General solution:

$$x = e^{-t} [C_1 \cos t + C_2 \sin t] + e^t [C_3 \cos t + C_4 \sin t]$$

Solve the following diff. eqs.

(1)  $\frac{d^4 y}{dx^4} - y = 0$

(2)  $(D^2 - 6D + 9)y = 0$

(3)  $\frac{d^3 y}{dx^3} - 13 \frac{dy}{dx} + 12y = 0$

(4)  $(D^3 + 1)y = 0$

(5)  $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 5y = 0$

Solution of non-homogeneous diff. eq.

Consider the diff. eq:

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = \phi(x)$$

$$\therefore (a_0 D^2 + a_1 D + a_2)y = \phi(x) \rightarrow (1)$$

$$\therefore f(D)y = \phi(x)$$

$$\text{where } f(D) = a_0 D^2 + a_1 D + a_2$$

Homogeneous part of this eq is given by

$$(a_0 D^2 + a_1 D + a_2)y \rightarrow (2)$$

$$\therefore f(D)y = 0$$

General solution of eq (2) is called complementary function (CF) and any solution of (1) is called a particular integral (PI)

### Theorem

Consider the diff. eq (1)

$$(a_0 D^2 + a_1 D + a_2) y = \phi(x) \longrightarrow (1)$$

It's homogeneous part is given by

$$(a_0 D^2 + a_1 D + a_2) y = 0 \longrightarrow (2)$$

General solution of eq (1) is given by

$$y = CF + PI$$

Proof Let  $y_g$  be the general soln of eq (2)

$$\text{i.e. } (a_0 D^2 + a_1 D + a_2) y_g = 0 \longrightarrow (3)$$

and  $y_p$  be a particular soln of eq (1)

$$\text{i.e. } (a_0 D^2 + a_1 D + a_2) y_p = \phi(x) \longrightarrow (4)$$

$$\text{Let } y = y_g + y_p \longrightarrow (3)$$

$$\begin{aligned} (a_0 D^2 + a_1 D + a_2) y &= (a_0 D^2 + a_1 D + a_2)(y_g + y_p) \\ &= (a_0 D^2 + a_1 D + a_2) y_g + (a_0 D^2 + a_1 D + a_2) y_p \\ &= 0 + \phi(x) \quad (\text{using (3) and (4)}) \\ &= \phi(x) \end{aligned}$$

This shows that  $y = y_g + y_p$  satisfies eq (1)

$\therefore$  It is a solution of eq (1).

Since  $y_g$  is the gen. soln of eq (2)

it contains two arbitrary constants

eq (3) gives the gen. soln of (1).

$$\text{i.e. } y = y_g + y_p = CF + PI \text{ is the gen. soln.}$$

## Special methods of finding Particular integral

consider the diff. eq

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = \phi(x)$$

$$\text{or } f(D)y = \phi(x) \quad \text{where } f(D) = a_0 D^2 + a_1 D + a_2$$

It's gen. soln is given by

$$y = CF + PI$$

Case (I)  $\phi(x) = e^{ax}$

$$D(e^{ax}) = a e^{ax}, \quad D^2(e^{ax}) = a^2 e^{ax}$$

$$\begin{aligned} \therefore f(D)(e^{ax}) &= (a_0 D^2 + a_1 D + a_2) e^{ax} \\ &= a_0 a^2 e^{ax} + a_1 a e^{ax} + a_2 e^{ax} \\ &= (a_0 a^2 + a_1 a + a_2) e^{ax} \\ &= f(a) e^{ax} \end{aligned}$$

Operating both sides by  $f(D)^{-1} = \frac{1}{f(D)}$ , we get

$$\frac{1}{f(D)} \cdot f(D) e^{ax} = \frac{1}{f(D)} f(a) e^{ax}$$

$$\text{or } e^{ax} = f(a) \cdot \frac{1}{f(D)} e^{ax}$$

$$\therefore \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax} \quad \text{provided } f(a) \neq 0.$$

$$\text{or } PI = \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax} \quad \text{if } f(a) \neq 0$$

Note  $f(a) = 0 \Rightarrow a$  is a root of the A.E.  
 $f(D) = 0$

If 'a' is a root of multiplicity  $r$  then

$$PI = \frac{x^r}{f^{(r)}(a)} e^{ax} \quad \text{where } f^{(r)}(a) = f^{(r)}(D) \text{ at } D=a$$

$$\text{when } r=1, \quad PI = \frac{x}{f'(a)} e^{ax}$$

$$\text{when } r=2 \quad PI = \frac{x^2}{f''(a)} e^{ax} \dots$$



Examples

(1) solve  $\frac{d^2 y}{dx^2} - 13 \frac{dy}{dx} + 12y = e^{-2x}$

Associated eq is given by

$$\frac{d^2 y}{dx^2} - 13 \frac{dy}{dx} + 12y = 0 \Rightarrow (D^2 - 13D + 12)y = 0$$

A.E. is given by  $D^2 - 13D + 12 = 0$

$$\therefore (D-1)(D-12) = 0 \Rightarrow D = 1, 12$$

$$\therefore CF = C_1 e^x + C_2 e^{12x}$$

$$PI = \frac{1}{f(D)} e^{-2x} = \frac{1}{D^2 - 13D + 12} e^{-2x} \quad (a = -2)$$

$$= \frac{1}{(-2)^2 - 13(-2) + 12} e^{-2x} \quad \left( \begin{array}{l} \text{Replacing} \\ D \text{ by } -2 \end{array} \right)$$

$$= \frac{1}{4 + 26 + 12} e^{-2x}$$

$$= \frac{1}{42} e^{-2x}$$

Gen. soln. is given by

$$y = CF + PI = C_1 e^x + C_2 e^{12x} + \frac{1}{42} e^{-2x}$$

(2) solve  $\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} - 10y = e^{2x}$

$$\therefore (D^2 + 3D - 10)y = e^{2x}$$

A.E. is  $D^2 + 3D - 10 = 0$

$$\therefore (D-2)(D+5) = 0 \Rightarrow D = 2, -5$$

$$CF = C_1 e^{2x} + C_2 e^{-5x}$$

$$PI = \frac{1}{f(D)} e^{2x} = \frac{1}{D^2 + 3D - 10} e^{2x} \quad \left( \begin{array}{l} a=2 \text{ which is} \\ \text{single root of A.E.} \end{array} \right)$$

$$= \frac{x}{f'(a)} e^{2x}$$

$$= \frac{x}{7} e^{2x}$$

$$\left| \begin{array}{l} f(D) = D^2 + 3D - 10 \\ f'(D) = 2D + 3 \\ f'(a) = 2(2) + 3 = 7 \end{array} \right|$$

$$\text{gen. soln. } y = CF + PI = C_1 e^{2x} + C_2 e^{-5x} + \frac{x}{7} e^{2x}$$



$$(3) \frac{d^2 y}{dx^2} - 12 \frac{dy}{dx} + 36y = e^{6x}$$

$$\text{or } (D^2 - 12D + 36)y = e^{6x}$$

$$\text{A.E. is } D^2 - 12D + 36 = 0$$

$$\text{or } (D-6)^2 = 0 \Rightarrow D = 6, 6$$

$$\text{CF} = (C_1 + C_2 x) e^{6x}$$

$$\text{PI} = \frac{1}{D^2 - 12D + 36} e^{6x}$$

$$= \frac{x^2}{f''(a)} e^{6x}$$

$$= \frac{x^2}{2} e^{6x}$$

( $a=6$  which is a double root of the A.E.)

$$\left| \begin{array}{l} f(D) = D^2 - 12D + 36 \\ f'(D) = 2D - 12 \\ f''(D) = 2 \therefore f''(6) = 2 \end{array} \right|$$

gen. soln is  $y = \text{CF} + \text{PI}$

$$= (C_1 + C_2 x) e^{6x} + \frac{x^2}{2} e^{6x}$$

Solve (1)  $\frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 5y = 2e^x + 10e^{-5x}$

(2)  $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 4y = 4e^x - 18e^{-x}$

(3)  $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^x + 4e^{3x}$

Answers

(1)  $(D^2 + 6D + 5)y = 2e^x + 10e^{-5x}$

A.E. is  $D^2 + 6D + 5 = 0$

or  $(D+1)(D+5) = 0 \Rightarrow D = -1, -5$

CF =  $C_1 e^{-x} + C_2 e^{-5x}$

PI =  $\frac{1}{D^2 + 6D + 5} (2e^x + 10e^{-5x})$

$= \frac{1}{D^2 + 6D + 5} (2e^x) + \frac{1}{D^2 + 6D + 5} (10e^{-5x})$

$= \text{PI}_1 + \text{PI}_2$

PI<sub>1</sub> =  $\frac{1}{D^2 + 6D + 5} (2e^x) = \frac{1}{1^2 + 6(1) + 5} (2e^x)$

( $a=1$  which is not a root of A.E.)  
 $\therefore$  Replace  $D$  by  $1$

$$i \quad PI_1 = \frac{1}{12} (2e^x) = \frac{1}{6} e^x$$

$$\begin{aligned}
 PI_2 &= \frac{1}{D^2 + bD + 5} (10e^{-5x}) \\
 &= \frac{x}{f'(a)} (10e^{-5x}) \quad \left( a = -5 \text{ which is a single root of the A.E.} \right) \\
 &= \frac{x}{-4} (10e^{-5x}) \\
 &= -\frac{5x}{2} e^{-5x}
 \end{aligned}
 \quad \left| \begin{array}{l} f(D) = D^2 + bD + 5 \\ f'(D) = 2D + b \\ f'(a) = 2(-5) + b = -4 \end{array} \right|$$

$$\text{gen. soln } y = CF + PI$$

$$y = CF + PI_1 + PI_2$$

$$i.e. \quad y = c_1 e^{\frac{x}{2}} + c_2 e^{-\frac{\sqrt{7}}{2}x} + \frac{1}{6} e^x - \frac{5x}{2} e^{-5x}$$

$$(2) \quad (D^2 - 3D + 4)y = 4e^x - 18e^{-x}$$

$$\text{A.E. is } D^2 - 3D + 4 = 0$$

$$D = \frac{3 \pm \sqrt{9 - 16}}{2} = \frac{3 \pm i\sqrt{7}}{2} \quad \left( \alpha = \frac{3}{2}, \beta = \frac{\sqrt{7}}{2} \right)$$

$$\begin{aligned}
 CF &= e^{\alpha x} [c_1 \cos \beta x + c_2 \sin \beta x] \\
 &= e^{\frac{3x}{2}} \left[ c_1 \cos \frac{\sqrt{7}}{2}x + c_2 \sin \frac{\sqrt{7}}{2}x \right]
 \end{aligned}$$

$$\begin{aligned}
 PI &= \frac{1}{D^2 - 3D + 4} (4e^x - 18e^{-x}) \\
 &= \frac{1}{D^2 - 3D + 4} (4e^x) + \frac{1}{D^2 - 3D + 4} (-18e^{-x}) \\
 &= \frac{1}{D^2 - 3D + 4} (4e^x) + \frac{1}{(-1)^2 - 3(-1) + 4} (-18e^{-x}) \\
 &= \frac{1}{2} (4e^x) + \frac{1}{8} (-18e^{-x}) \\
 &= 2e^x - \frac{9}{2} e^{-x}
 \end{aligned}$$

$$\text{gen. soln } y = CF + PI$$

$$= e^{\frac{3x}{2}} \left[ c_1 \cos \frac{\sqrt{7}}{2}x + c_2 \sin \frac{\sqrt{7}}{2}x \right] + 2e^x - \frac{9}{2} e^{-x}$$

$$(3) (D^2 - 3D + 2)y = e^x + 4e^{3x}$$

$$\text{A.E. is } D^2 - 3D + 2 = 0 \Rightarrow (D-1)(D-2) = 0$$

$$\therefore D = 1, 2$$

$$CF = C_1 e^x + C_2 e^{2x}$$

$$PI = \frac{1}{D^2 - 3D + 2} (e^x + 4e^{3x})$$

$$= \frac{1}{D^2 - 3D + 2} (e^x) + \frac{1}{D^2 - 3D + 2} (4e^{3x})$$

$$= PI_1 + PI_2$$

$$PI_1 = \frac{1}{D^2 - 3D + 2} (e^x)$$

(Here  $a=1$  which is a single root of the A.E.)

$$= \frac{x}{f'(a)} e^{ax}$$

$$= \frac{x}{-1} e^x = -x e^x$$

$$\left( \begin{array}{l} f(D) = D^2 - 3D + 2 \\ f'(D) = 2D - 3 \\ f'(a) = 2(1) - 3 = -1 \end{array} \right)$$

$$PI_2 = \frac{1}{D^2 - 3D + 2} (4e^{3x}) \quad \left( \text{Here } a=3 \text{ which is not a root of the A.E.} \right)$$

$$= \frac{1}{3^2 - 3(3) + 2} (4e^{3x}) = \frac{1}{2} (4e^{3x}) = 2e^{3x}$$

$$\text{Gen. Soln } y = CF + PI = C_1 e^x + C_2 e^{2x} - x e^x + 2e^{3x}$$

Solve the following diff. eqs

$$(1) (D^2 - 5D + 6)y = e^{5x}$$

$$(2) (D^2 + 5D + 7)y = e^{-x}$$

$$(2) (D^2 + 4D + 4)y = e^{-3x}$$

$$(4) \frac{d^2 y}{dx^2} - 9y = 5e^{4x} + 2e^{3x}$$

$$(5) (D^2 - 6D + 8)y = (e^{2x} - 1)^2$$



Case II when  $\phi(x) = \sin ax$  or  $\cos ax$

$$D(\sin ax) = a \cos ax, \quad D^2(\sin ax) = -a^2 \sin ax$$

$$\text{i.e. } D^2(\sin ax) = (-a^2) \sin ax$$

$$\begin{aligned} D^4(\sin ax) &= D^2(D^2 \sin ax) = D^2(-a^2 \sin ax) \\ &= (-a^2)(-a^2 \sin ax) \\ &= (-a^2)^2 \sin ax \end{aligned}$$

$$\text{In general } D^{2n}(\sin ax) = (-a^2)^n \sin ax$$

$$\therefore f(D^2) \sin ax = f(-a^2) \sin ax$$

$$\frac{1}{f(D^2)} f(D^2) \sin ax = \frac{1}{f(D^2)} f(-a^2) \sin ax$$

$$\text{i.e. } \sin ax = \frac{1}{f(D^2)} f(-a^2) \sin ax$$

$$\therefore \frac{1}{f(D^2)} \sin ax = \frac{1}{f(-a^2)} \sin ax \text{ provided } f(-a^2) \neq 0$$

Remark If  $f(-a^2) = 0$ , the above method fails.

However when  $f(-a^2) = 0 \Rightarrow D^2 + a^2$  is a factor of  $f(D^2)$  we evaluate

$$\frac{1}{D^2 + a^2} (\sin ax) = \frac{1}{D^2 + a^2} (\text{Im. part of } e^{iax})$$

$$(\because e^{iax} = \cos ax + i \sin ax)$$

$$= \text{Im. p. of } \frac{1}{D^2 + a^2} (e^{iax})$$

$$= \text{Im. p. of } \frac{x}{2ia} e^{iax}$$

$$\left( \begin{array}{l} F(D) = D^2 + a^2 \\ F'(D) = 2D \\ F'(ia) = 2ia \end{array} \right)$$

$$= \text{Im. p. of } \frac{ix}{-2a} (\cos ax + i \sin ax)$$

$$= \text{Im. p. of } \frac{x}{2a} (-i \cos ax + \sin ax)$$

$$= \underline{\underline{-\frac{x}{2a} \cos ax}}$$



In a similar manner we can obtain that

$$\frac{1}{f(D^2)} \cos ax = \frac{1}{f(-a^2)} \cos ax \text{ provided } f(-a^2) \neq 0$$

and if  $f(-a^2) = 0 \Rightarrow D^2 + a^2$  is a factor of  $f(D^2)$

and we obtain

$$\begin{aligned} \frac{1}{D^2 + a^2} \cos ax &= \frac{1}{D^2 + a^2} (\text{Re. part of } e^{iax}) \\ &= \text{Real part of } \frac{1}{D^2 + a^2} (e^{iax}) \\ &= \frac{x}{2a} \sin ax \end{aligned}$$

Solve the following diff. eqs.

- (1)  $(D^2 + 4)y = \cos 3x$       (2)  $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = e^{3x} + \cos 5x$   
 (3)  $(D^2 + 25)y = 2 \sin 5x$       (4)  $(D^2 - 2D - 8)y = 4 \cos 2x + e^{4x}$   
 (5)  $(D^2 + 16)y = e^{-3x} - \cos 4x$

(1)  $(D^2 + 4)y = \cos 3x$

To find CF we solve  $(D^2 + 4)y = 0$

$$D^2 + 4 = 0 \Rightarrow D^2 = -4 \Rightarrow D = \pm 2i$$

$$CF = C_1 \cos 2x + C_2 \sin 2x$$

$$PI = \frac{1}{D^2 + 4} \cos 3x = \frac{1}{-3^2 + 4} \cos 3x = -\frac{1}{5} \cos 3x$$

Solution is given by

$$y = C_1 \cos 2x + C_2 \sin 2x - \frac{1}{5} \cos 3x$$

(2)  $(D^2 + 4D + 4)y = e^{3x} + \cos 5x$

To find CF, solve  $(D^2 + 4D + 4)y = 0$

$$\begin{aligned} \text{The A.E is } D^2 + 4D + 4 &= 0 \Rightarrow (m+2)^2 = 0 \\ \Rightarrow (m+2)(m+2) &= 0 \Rightarrow m = -2, -2 \end{aligned}$$

$$CF = (C_1 x + C_2) e^{-2x}$$

$$PI = \frac{1}{D^2 + 4D + 4} (e^{3x} + \cos 5x)$$

$$= \frac{1}{D^2 + 4D + 4} (e^{3x}) + \frac{1}{D^2 + 4D + 4} (\cos 5x) = PI_1 + PI_2$$

$$PI_1 = \frac{1}{3^2 + 4(3) + 4} e^{3x} = \frac{1}{25} e^{3x}$$

$$PI_2 = \frac{1}{D^2 + 4D + 4} (\cos 5x)$$

$$= \frac{1}{-5^2 + 4D + 4} (\cos 5x) \quad (\text{Replacing } D^2 \text{ by } -5^2)$$

$$= \frac{1}{4D - 21} (\cos 5x)$$

$$= \frac{4D + 21}{(4D - 21)(4D + 21)} (\cos 5x)$$

$$= \frac{4D + 21}{16D^2 - 441} (\cos 5x)$$

$$= \frac{4D + 21}{16(-5^2) - 441} (\cos 5x) \quad (\text{Replacing } D^2 \text{ by } -5^2)$$

$$= \frac{4D + 21}{-841} (\cos 5x) = \frac{-1}{841} (4(-5 \sin 5x) + 21 \cos 5x)$$

$$= \frac{-1}{841} (21 \cos 5x - 20 \sin 5x)$$

Hence gen. soln is given by

$$y = (c_1 + c_2 x) e^{-2x} + \frac{1}{25} e^{3x} - \frac{1}{841} (21 \cos 5x - 20 \sin 5x)$$

$$(3) (D^2 + 25)y = 25 \sin 5x$$

To find CF solve  $(D^2 + 25)y = 0$

$$\text{A.E. is } D^2 + 25 = 0 \Rightarrow D^2 = -25 \Rightarrow D = \pm 5i$$

$$CF = c_1 \cos 5x + c_2 \sin 5x$$

$$P.I. = \frac{1}{D^2 + 25} (\sin 5x) =$$

$$= -\frac{x}{2 \times 5} \cos 5x$$

$$= -\frac{x}{10} \cos 5x$$

$$\frac{1}{D^2 + a^2} (\sin ax) = -\frac{x}{2a} \cos ax$$

gen. soln is  $y = CF + PI$

$$\text{ie } y = C_1 \cos 5x + C_2 \sin 5x - \frac{x}{10} \cos 5x$$

$$(4) (D^2 - 2D - 8)y = 4 \cos 2x + e^{4x}$$

CF is got by solving  $(D^2 - 2D - 8)y = 0$

$$A.E. \text{ is } D^2 - 2D - 8 = 0 \Rightarrow (D+2)(D-4)$$

$$\Rightarrow D = -2, 4$$

$$CF = C_1 e^{-2x} + C_2 e^{4x}$$

$$PI = \frac{1}{D^2 - 2D - 8} (4 \cos 2x + e^{4x})$$

$$= \frac{1}{D^2 - 2D - 8} (4 \cos 2x) + \frac{1}{D^2 - 2D - 8} (e^{4x})$$

$$P.I_1 = \frac{1}{D^2 - 2D - 8} (4 \cos 2x)$$

$$= 4 \cdot \frac{1}{-2^2 - 2D - 8} (\cos 2x) = -\frac{4}{2D + 12} \cos 2x$$

$$= -\frac{4}{2(D+6)} \cos 2x = -\frac{2}{(D+6)} \cos 2x$$

$$= -\frac{2(D-6)}{D^2 - 36} \cos 2x = \frac{-2(D-6)}{-2^2 - 36} \cos 2x$$

$$= \frac{-2}{-40} (D-6) \cos 2x = \frac{-1}{20} (-2 \sin 2x - 6 \cos 2x)$$

$$PI_2 = \frac{1}{D^2 - 2D - 8} (e^{4x}) = \frac{x}{-6^2 - 8} e^{4x}$$

$$PI = \frac{1}{20} (2 \sin 2x + 6 \cos 2x) + \frac{x e^{4x}}{6}$$

gen. soln

$$y = C_1 e^{-2x} + C_2 e^{4x} + \frac{1}{20} (2 \sin 2x + 6 \cos 2x) + \frac{x e^{4x}}{6}$$

$$\begin{cases} f(D) = D^2 - 2D - 8 \\ f'(D) = 2D - 2 \\ f'(4) = 6 \end{cases}$$

(1) Solve  $\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = \sin 2x$

(2)  $(D^2 + 4D + 4)y = \cos x$

(3)  $(D^2 - 4D - 5)y = e^{3x} + 4 \cos 3x$

(4)  $(D^2 + 9)y = \cos 3x$



① Solve  $\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = \sin 2x$

(2)  $(D^2 + 4D + 4)y = \cos x$

(3)  $(D^2 - 4D - 5)y = e^{3x} + 4 \cos 3x$

(4)  $(D^2 + 9)y = \cos 3x$

(5)  $(D^2 + 1)y = \sin x \sin 2x$  (6)  $(2D^2 - 5D + 3)y = \cos 3x \cos 2x$

Case III when  $\phi(x) = x^m$ ,  $m$  being a +ve integer

Here we have to evaluate  $\frac{1}{f(D)} x^m$ . For that expand  $\frac{1}{f(D)} = [f(D)]^{-1}$  in ascending powers of  $D$  and operate on  $x^m$ . we use the following results for expansion.

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots \quad (1)$$

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots \quad (2)$$

$$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots \quad (3)$$

$$(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots \quad (4)$$

Solve the following diff. eqs.

(1)  $(D^2 + 2D + 1)y = 2x + x^2$

(2)  $(D^2 + D + 1)y = x^2$

(3)  $(D^2 + 25)y = x^2 + x$

Answers

(1)  $(D^2 + 2D + 1)y = 2x + x^2$

CF is got by solving  $(D^2 + 2D + 1)y = 0$

A.E. is  $D^2 + 2D + 1 = 0 \Rightarrow (D+1)^2 = 0$

$\Rightarrow (D+1)(D+1) = 0 \Rightarrow D = -1, -1$

CF =  $(C_1 + C_2 x) e^{-x}$

PI =  $\frac{1}{D^2 + 2D + 1} (2x + x^2) = \frac{1}{(D+1)^2} (2x + x^2)$

$= (1+D)^{-2} (2x + x^2)$

$= (1 - 2D + 3D^2 - 4D^3 + \dots)(2x + x^2)$  using series (4)

$$\begin{aligned}
 \text{16. PI} &= (2x + x^2) - 2D(2x + x^2) + 3D^2(2x + x^2) - \dots \\
 &= (2x + x^2) - 2(2 + 2x) + 3(2) + 0 \\
 &= x^2 + 2x - 4 - 4x + 6 \\
 &= x^2 - 2x + 2
 \end{aligned}$$

$$\begin{aligned}
 \text{Gen. soln. } y &= CF + PI \\
 &= (C_1 + C_2 x) e^{-x} + x^2 - 2x + 2
 \end{aligned}$$

$$(2) (D^2 + D + 1)y = x^2$$

CF is got by solving  $(D^2 + D + 1)y = 0$

$$\text{A.E. is } D^2 + D + 1 = 0$$

$$\begin{aligned}
 \text{Solving } D &= \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{-3}}{2} \quad \alpha = -\frac{1}{2}, \beta = \frac{\sqrt{3}}{2} \\
 &= \frac{-1 \pm i\sqrt{3}}{2}
 \end{aligned}$$

$$CF = e^{-\frac{x}{2}} \left[ C_1 \cos \frac{\sqrt{3}}{2} x + C_2 \sin \frac{\sqrt{3}}{2} x \right]$$

$$PI = \frac{1}{D^2 + D + 1} (x^2)$$

$$= (1 + D + D^2)^{-1} (x^2)$$

$$= x^2 - (D + D^2)(x^2) + (D + D^2)^2(x^2) - \dots$$

$$= x^2 - (2x + 2) + (D^2 + 2D^3 + D^4)(x^2) - \dots$$

$$= x^2 - (2x + 2) + (2 + 0 + 0) - 0$$

$$= x^2 - 2x - 2 + 2$$

$$= x^2 - 2x$$

$$\text{Gen. solution: } y = e^{-\frac{x}{2}} \left[ C_1 \cos \frac{\sqrt{3}}{2} x + C_2 \sin \frac{\sqrt{3}}{2} x \right] + x^2 - 2x$$

$$(3) (D^2 + 25)y = x^2 + x$$

CF is got by solving  $(D^2 + 25)y = 0$

$$\text{A.E. is } D^2 + 25 = 0 \Rightarrow D^2 = -25 \Rightarrow D = \pm 5i$$

$$CF = C_1 \cos 5x + C_2 \sin 5x \quad (x^2 + x)$$

$$\begin{aligned}
 PI &= \frac{1}{D^2 + 25} (x^2 + x) = \frac{1}{25 \left( \frac{D^2}{25} + 1 \right)} (x^2 + x) \\
 &= \frac{1}{25} \left( 1 + \frac{D^2}{25} \right)^{-1} (x^2 + x)
 \end{aligned}$$

$$P.E = \frac{1}{25} \left( 1 - \frac{D^2}{25} + \frac{D^4}{625} - \dots \right) (x^2 + x)$$

$$= \frac{1}{25} \left[ (x^2 + x) - \frac{D^2}{25} (x^2 + x) + \frac{D^4}{625} (x^2 + x) - \dots \right]$$

$$= \frac{1}{25} \left[ x^2 + x - \frac{2}{25} + 0 \right] = \frac{1}{625} (25x^2 + 25x - 2)$$

$$\text{Gen. soln: } y = C_1 \cos 5x + C_2 \sin 5x + \frac{1}{625} (25x^2 + 25x - 2)$$

Solve (1)  $(D^2 - 2D)y = 2 + 3x$

(2)  $(D^2 - 1)y = 3 + 2x^2$  (3)  $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = x^2 + e^{4x}$

(4)  $(D^2 + 4D + 5)y = e^{2x} + \cos x + x^2$

① Solve  $\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = \sin 2x$

(2)  $(D^2 + 4D + 4)y = \cos x$

(3)  $(D^2 - 4D - 5)y = e^{3x} + 4 \cos 3x$

(4)  $(D^2 + 9)y = \cos 3x$

(5)  $(D^2 + 1)y = \sin x \sin 2x$  (6)  $(2D^2 - 5D + 3)y = \cos 3x \cos 2x$

Case III when  $\phi(x) = x^m$ ,  $m$  being a +ve integer

Here we have to evaluate  $\frac{1}{f(D)} x^m$ . For that expand  $\frac{1}{f(D)} = [f(D)]^{-1}$  in ascending powers of  $D$  and operate on  $x^m$ . we use the following results for expansion.

$(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$  (1)

$(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$  (2)

$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$  (3)

$(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$  (4)

solve the following diff. eqs.

(1)  $(D^2 + 2D + 1)y = 2x + x^2$

(2)  $(D^2 + D + 1)y = x^2$

(3)  $(D^2 + 25)y = x^2 + x$

Answers

(1)  $(D^2 + 2D + 1)y = 2x + x^2$

CF is got by solving  $(D^2 + 2D + 1)y = 0$

A.E. is  $D^2 + 2D + 1 = 0 \Rightarrow (D+1)^2 = 0$

$\Rightarrow (D+1)(D+1) = 0 \Rightarrow D = -1, -1$

CF =  $(C_1 + C_2 x) e^{-x}$

PI =  $\frac{1}{D^2 + 2D + 1} (2x + x^2) = \frac{1}{(D+1)^2} (2x + x^2)$

$= (1+D)^{-2} (2x + x^2)$

$= (1 - 2D + 3D^2 - 4D^3 + \dots)(2x + x^2)$  using series (4)



$$\begin{aligned}
 \text{PI} &= (2x + x^2) - 2D(2x + x^2) + 3D^2(2x + x^2) - \dots \\
 &= (2x + x^2) - 2(2 + 2x) + 3(2) + 0 \\
 &= x^2 + 2x - 4 - 4x + 6 \\
 &= x^2 - 2x + 2
 \end{aligned}$$

$$\begin{aligned}
 \text{Gen. soln. } y &= CF + PI \\
 &= (C_1 + C_2 x) e^{-x} + x^2 - 2x + 2
 \end{aligned}$$

$$(2) (D^2 + D + 1)y = x^2$$

CF is got by solving  $(D^2 + D + 1)y = 0$

$$\text{A.E. is } D^2 + D + 1 = 0$$

$$\begin{aligned}
 \text{Solving } D &= \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{-3}}{2} \quad \alpha = -\frac{1}{2}, \beta = \frac{\sqrt{3}}{2} \\
 &= \frac{-1 \pm i\sqrt{3}}{2}
 \end{aligned}$$

$$CF = e^{-\frac{x}{2}} \left[ C_1 \cos \frac{\sqrt{3}}{2} x + C_2 \sin \frac{\sqrt{3}}{2} x \right]$$

$$PI = \frac{1}{D^2 + D + 1} (x^2)$$

$$= (1 + D + D^2)^{-1} (x^2)$$

$$= x^2 - (D + D^2)(x^2) + (D + D^2)^2(x^2) - \dots$$

$$= x^2 - (2x + 2) + (D^2 + 2D^3 + D^4)(x^2) - \dots$$

$$= x^2 - (2x + 2) + (2 + 0 + 0) - 0$$

$$= x^2 - 2x - 2 + 2$$

$$= x^2 - 2x$$

$$\text{Gen. solution: } y = e^{-\frac{x}{2}} \left[ C_1 \cos \frac{\sqrt{3}}{2} x + C_2 \sin \frac{\sqrt{3}}{2} x \right] + x^2 - 2x$$

$$(3) (D^2 + 25)y = x^2 + x$$

CF is got by solving  $(D^2 + 25)y = 0$

$$\text{A.E. is } D^2 + 25 = 0 \Rightarrow D = \pm 5i$$

$$CF = C_1 \cos 5x + C_2 \sin 5x$$

$$\begin{aligned}
 PI &= \frac{1}{D^2 + 25} (x^2 + x) = \frac{1}{25 \left( \frac{D^2}{25} + 1 \right)} (x^2 + x) \\
 &= \frac{1}{25} \left( 1 + \frac{D^2}{25} \right)^{-1} (x^2 + x)
 \end{aligned}$$

$$PI = \frac{1}{25} \left( 1 - \frac{D^2}{25} + \frac{D^4}{625} - \dots \right) (x^2 + x)$$

$$= \frac{1}{25} \left[ (x^2 + x) - \frac{D^2}{25} (x^2 + x) + \frac{D^4}{625} (x^2 + x) - \dots \right]$$

$$= \frac{1}{25} \left[ x^2 + x - \frac{2}{25} + 0 \right] = \frac{1}{625} (25x^2 + 25x - 2)$$

$$\text{Gen. soln: } y = C_1 \cos 5x + C_2 \sin 5x + \frac{1}{625} (25x^2 + 25x - 2)$$

Solve (1)  $(D^2 - 2D)y = 2 + 3x$

(2)  $(D^2 - 1)y = 3 + 2x^2$

(3)  $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = x^2 + e^{4x}$

(4)  $(D^2 + 4D + 5)y = e^{2x} + \cos x + x^2$

Case IV when  $\phi(x) = e^{ax} \cdot v$  where  $v$  is a function of  $\sin ax, \cos ax$  or  $x^m$

$$D(e^{ax} \cdot v) = e^{ax} D(v) + a e^{ax} v$$

$$= e^{ax} (D + a)v$$

$$D^2(e^{ax} \cdot v) = D[e^{ax} (D + a)v]$$

$$= e^{ax} \cdot D(D + a)v + (D + a)v(a e^{ax})$$

$$= e^{ax} [D(D + a) + a(D + a)]v$$

$$= e^{ax} [(D + a)(D + a)]v$$

$$= e^{ax} (D + a)^2 v$$

$$\therefore f(D)(e^{ax} \cdot v) = e^{ax} f(D + a)v$$

$$\text{This shows that } \frac{1}{f(D)}(e^{ax} \cdot v) = e^{ax} \frac{1}{f(D + a)}v$$

Problems Solve  $(D^2 + 3D + 2)y = e^{2x} \sin x$

To find CF solve  $(D^2 + 3D + 2)y = 0$

A.E. is  $D^2 + 3D + 2 = 0$

$\therefore D = -1, -2$

CF =  $C_1 e^{-x} + C_2 e^{-2x}$

$$P.I = \frac{1}{D^2+3D+2} (e^{2x} \sin x)$$

$$= e^{2x} \cdot \frac{1}{(D+2)^2+3(D+2)+2} (\sin x) \quad (\text{Replace } D \text{ by } D+2)$$

$$= e^{2x} \cdot \frac{1}{D^2+4D+4+3D+6+2} (\sin x)$$

$$= e^{2x} \cdot \frac{1}{D^2+7D+12} (\sin x)$$

$$= e^{2x} \cdot \frac{1}{-1^2+7D+12} (\sin x)$$

(a=1  
Replace  $D^2$  by  $-1^2$ )

$$= e^{2x} \cdot \frac{1}{7D+11} (\sin x)$$

$$= e^{2x} \cdot \frac{7D-11}{49D^2-121} (\sin x)$$

(Replacing  $D^2$  by  $-1^2$ )

$$= e^{2x} \cdot \frac{7D-11}{49(-1^2)-121} (\sin x)$$

$$= e^{2x} \cdot \frac{7D-11}{-170} (\sin x)$$

$$= e^{2x} \left( -\frac{1}{170} \right) (7 \cos x - 11 \sin x)$$

$$= \frac{1}{170} e^{2x} (11 \sin x - 7 \cos x)$$

Gen. Soln.  $y = CF + P.I$

$$= C_1 e^{-x} + C_2 e^{-2x} + \frac{1}{170} e^{2x} (11 \sin x - 7 \cos x)$$

(2) solve  $(D^2-2D+2)y = e^x x^3$

To find CF solve  $(D^2-2D+2)y = 0$

The auxillary eq is  $D^2-2D+2=0$

$$D = \frac{2 \pm \sqrt{4-8}}{2} = \frac{2 \pm 2i}{2} = 1 \pm i$$

$$CF = e^x (C_1 \cos x + C_2 \sin x)$$

$$P.I = \frac{1}{D^2-2D+2} (e^x \cdot x^3) = e^x \cdot \frac{1}{(D+1)^2-2(D+1)+2} (x^3)$$

$$= e^x \frac{1}{D^2 + 2D + 1 - 2D - 2 + 2} (x^3)$$

$$= e^x \frac{1}{D^2 + 1} (x^3)$$

$$= e^x (1 + D^2)^{-1} (x^3)$$

$$= e^x [1 - D^2 + D^4 - D^6 + \dots] (x^3)$$

$$= e^x (x^3 - 6x + 0) = e^x (x^3 - 6x)$$

$\therefore$  gen. soln

$$y = CF + PI = e^x [C_1 \cos x + C_2 \sin x] + e^x (x^3 - 6x)$$

(3) Solve  $\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = e^{-2x} \sin x$

CF is got by solving  $(D^2 + 5D + 6)y = 0$

A.E. is given by  $D^2 + 5D + 6 = 0$

$$\text{i.e. } (D+2)(D+3) = 0 \Rightarrow D = -2, -3$$

$$CF = C_1 e^{-2x} + C_2 e^{-3x}$$

$$PI = \frac{1}{D^2 + 5D + 6} (e^{-2x} \sin x)$$

$$= e^{-2x} \cdot \frac{1}{(D-2)^2 + 5(D-2) + 6} (\sin x)$$

$$= e^{-2x} \cdot \frac{1}{D^2 - 4D + 4 + 5D - 10 + 6} (\sin x)$$

$$= e^{-2x} \cdot \frac{1}{D^2 + D} (\sin x)$$

$$= e^{-2x} \cdot \frac{1}{-1^2 + D} (\sin x) \quad \left( \begin{array}{l} a = 1 \\ \text{Replacing } D^2 \text{ by } -1^2 \end{array} \right)$$

$$= e^{-2x} \cdot \frac{1}{D-1} (\sin x)$$

$$= e^{-2x} \cdot \frac{D+1}{D^2-1} (\sin x) = e^{-2x} \cdot \frac{(D+1)}{-2} (\sin x)$$

$$= \frac{e^{-2x}}{-2} (\cos x + \sin x)$$



Gen. soln:  $y = CF + PI$

$$\text{ie } y = c_1 e^{-2x} + c_2 e^{-3x} \rightarrow \frac{e^{-2x}}{2} (\sin x + \cos x)$$

(4) Solve  $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = x e^x \sin x$

CF is got by solving  $(D^2 - 2D + 1)y = 0$

A.E is  $D^2 - 2D + 1 = 0 \Rightarrow D = 1, 1$

CF =  $(c_1 + c_2 x) e^x$ .

PI =  $\frac{1}{D^2 - 2D + 1} (x e^x \sin x)$

=  $\frac{1}{(D-1)^2} [e^x (x \sin x)]$

=  $e^x \cdot \frac{1}{(D+1-1)^2} (x \sin x)$

=  $e^x \cdot \frac{1}{D^2} (x \sin x)$

$\left( \frac{1}{D} \equiv \int \right)$

=  $e^x \cdot \frac{1}{D} \int x \sin x$

=  $e^x \frac{1}{D} [x(\cos x) - 1(-\sin x)]$  (Applying Integration by Parts)

=  $e^x \int (x \cos x + \sin x) dx$   $[ \int u v dx = u v_1 - u' v_2 ]$

=  $e^x \left[ \int x \cos x dx + \int \sin x dx \right]$

=  $e^x [-x \sin x + 1(-\cos x) - \cos x]$

=  $e^x [-x \sin x - 2 \cos x]$

=  $-e^x [x \sin x + 2 \cos x]$

Gen. soln:  $y = (c_1 + c_2 x) e^x - e^x [x \sin x + 2 \cos x]$   
 $= e^x [c_1 + c_2 x - (x \sin x + 2 \cos x)]$

## Cauchy - Euler Equation.

A diff. eq. of the form

$$a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = \phi(x)$$

where  $a_0, a_1, a_2, \dots, a_n$  are constants is called Cauchy - Euler diff. eq. This eq can be transformed into a linear diff. eq with constant coeffs by changing the independent variable  $x$  to  $z$  by substituting  $x = e^z$  i.e.  $z = \log x \therefore \frac{dz}{dx} = \frac{1}{x}$

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = \frac{1}{x} \frac{dy}{dz} \rightarrow (1)$$

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left[ \frac{1}{x} \cdot \frac{dy}{dz} \right] \\ &= \frac{1}{x} \cdot \frac{d}{dx} \left( \frac{dy}{dz} \right) + \frac{dy}{dz} \frac{d}{dx} \left( \frac{1}{x} \right) \\ &= \frac{1}{x} \frac{d}{dz} \left( \frac{dy}{dz} \right) \cdot \frac{dz}{dx} + \frac{dy}{dz} \left( -\frac{1}{x^2} \right) \\ &= \frac{1}{x} \frac{d^2 y}{dz^2} \cdot \frac{1}{x} - \frac{1}{x^2} \frac{dy}{dz} \\ &= \frac{1}{x^2} \left( \frac{d^2 y}{dz^2} - \frac{dy}{dz} \right) \end{aligned} \rightarrow (2)$$

Writing  $D$  for  $\frac{d}{dz}$  and  $D^2$  for  $\frac{d^2}{dz^2}$ , above

results may be written as

$$\frac{dy}{dx} = \frac{1}{x} D y \quad \text{i.e.} \quad x \frac{dy}{dx} = D y,$$

$$\frac{d^2 y}{dx^2} = \frac{1}{x^2} (D^2 - 1) y \quad \text{i.e.} \quad x^2 \frac{d^2 y}{dx^2} = D(D-1) y$$

$$\text{Similarly } x^3 \frac{d^3 y}{dx^3} = D(D-1)(D-2) y$$

$$\text{In general } x^n \frac{d^n y}{dx^n} = D(D-1)(D-2) \dots (D-n+1) y$$

When above substitutions are made we get a

linear diff eq. with constant coeffs.

Solve (1)  $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = x^3$

(2)  $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = \log x + x$

(3)  $x^3 \frac{d^3y}{dx^3} - 3x^2 \frac{d^2y}{dx^2} + 6x \frac{dy}{dx} - 6y = 0$

(4)  $x^3 \frac{d^3y}{dx^3} - x^2 \frac{d^2y}{dx^2} - 6x \frac{dy}{dx} + 18y = 0$

(1)  $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = x^3 \rightarrow (1)$

put  $x = e^z$  i.e.  $z = \log x$

$\therefore x \frac{dy}{dx} = Dy$ ,  $x^2 \frac{d^2y}{dx^2} = D(D-1)y$  where  $D = \frac{d}{dz}$

The transformed eq is

$$D(D-1)y - 2Dy + 2y = (e^z)^3$$

$$\text{i.e. } (D^2 - D - 2D + 2)y = e^{3z}$$

$$\text{i.e. } (D^2 - 3D + 2)y = e^{3z} \rightarrow (2)$$

This is a linear diff. eq with const coeffs

To find CF solve  $(D^2 - 3D + 2)y = 0$

A.E. is  $D^2 - 3D + 2 = 0$  i.e.  $(D-1)(D-2) = 0$

$\therefore D = 1, 2$  CF =  $c_1 e^z + c_2 e^{2z}$

PI =  $\frac{1}{D^2 - 3D + 2} (e^{3z}) = \frac{1}{3^2 - 3(3) + 2} e^{3z} = \frac{1}{2} e^{3z}$

gen. soln of eq (2) is given by

$$y = c_1 e^z + c_2 e^{2z} + \frac{1}{2} e^{3z}$$

i.e.  $y = c_1 x + c_2 x^2 + \frac{1}{2} x^3$

is the required soln

$$\left( \begin{array}{l} \because e^z = x \\ e^{2z} = (e^z)^2 = x^2 \\ e^{3z} = (e^z)^3 = x^3 \end{array} \right)$$

$$(2) \quad x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = \log x + x \rightarrow (1)$$

Put  $x = e^z$  then  $z = \log x$

$$x \frac{dy}{dx} = Dy, \quad x^2 \frac{d^2 y}{dx^2} = D(D-1)y, \quad \text{where } D \equiv \frac{d}{dz}$$

$\therefore$  The transformed eq is

$$D(D-1)y + Dy + y = z + e^z$$

$$\text{or } (D^2 - D + D + 1)y = z + e^z$$

$$\text{or } (D^2 + 1)y = z + e^z \rightarrow (2)$$

To find CF, solve  $(D^2 + 1)y = 0$

A.E. is given by  $D^2 + 1 = 0$ ,  $D = \pm i$

$$CF = C_1 \cos z + C_2 \sin z$$

$$PI = \frac{1}{D^2 + 1} (z + e^z) = \frac{1}{D^2 + 1} (z) + \frac{1}{D^2 + 1} (e^z)$$

$$= PI_1 + PI_2$$

$$PI_1 = \frac{1}{D^2 + 1} (z) = (1 + D^2)^{-1} z \\ = (1 - D^2 + D^4 - \dots) z \\ = z - 0 = z$$

$$PI_2 = \frac{1}{D^2 + 1} (e^z) = \frac{1}{2} e^z$$

$\therefore$  General soln of eq(2) is

$$y = C_1 \cos z + C_2 \sin z + z + \frac{1}{2} e^z$$

$$\text{or } y = C_1 \cos(\log x) + C_2 \sin(\log x) + \log x + \frac{x}{2}$$

$$(3) \quad x^3 \frac{d^3 y}{dx^3} - 3x^2 \frac{d^2 y}{dx^2} + 6x \frac{dy}{dx} - 6y = 0 \rightarrow (1)$$

Put  $x = e^z$ , then  $z = \log x$

$$x \frac{dy}{dx} = Dy, \quad x^2 \frac{d^2 y}{dx^2} = D(D-1)y$$

$$x^3 \frac{d^3 y}{dx^3} = D(D-1)(D-2)y \quad \text{where } D = \frac{d}{dz}$$



The transformed eq is

$$D(D-1)(D-2)y - 3D(D-1)y + 6Dy - 6y = 0$$

$$\text{i.e. } (D(D^2 - 3D + 2) - 3D(D-1) + 6D - 6)y = 0$$

$$\text{i.e. } (D^3 - 3D^2 + 2D - 3D^2 + 3D + 6D - 6)y = 0$$

$$\text{i.e. } (D^3 - 6D^2 + 11D - 6)y = 0 \quad \longrightarrow (2)$$

A.E. is given by  $D^3 - 6D^2 + 11D - 6 = 0$

$D=1$  is a solution of this cubic equation

Dividing  $D^3 - 6D^2 + 11D - 6$  by  $D-1$ ,

$$D^3 - 6D^2 + 11D - 6 = (D-1)(D^2 - 5D + 6)$$

$$\therefore (D-1)(D-2)(D-3) = 0$$

$$D = 1, 2, 3$$

Soln of eq (2) is given by

$$y = C_1 e^x + C_2 e^{2x} + C_3 e^{3x}$$

i.e.  $y = C_1 x + C_2 x^2 + C_3 x^3$  is the required soln.

$$\begin{array}{r} D-1 \ ) \ D^3 - 6D^2 + 11D - 6 \quad (D^2 - 5D + 6) \\ \underline{D^3 - D^2} \phantom{+ 11D - 6} \\ -5D^2 + 11D \phantom{- 6} \\ \underline{-5D^2 + 5D} \phantom{- 6} \\ 6D - 6 \\ \underline{6D - 6} \\ 0 \end{array}$$

Solve (1)  $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - 4y = x$

(2)  $x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} - 20y = x^4$

(3)  $x^3 \frac{d^3 y}{dx^3} - x^2 \frac{d^2 y}{dx^2} - 6x \frac{dy}{dx} + 18y = 0$

## Simultaneous Differential Equations

consider two diff. eqs of the form

$$\frac{dx}{dt} + a_1x + a_2y = f_1(t)$$

$$\frac{dy}{dt} + b_1x + b_2y = f_2(t)$$

Put  $D \equiv \frac{d}{dt}$

$\therefore$  we get 
$$\begin{aligned} Dx + a_1x + a_2y &= f_1(t) \\ Dy + b_1x + b_2y &= f_2(t) \end{aligned}$$

i.  $(D + a_1)x + a_2y = f_1(t) \rightarrow (1)$

$b_1x + (D + b_2)y = f_2(t) \rightarrow (2)$

Eliminate one of the variables  $x$  (or  $y$ ), then we get a second diff. eq in  $y$  and  $t$  (or  $x$  and  $t$ ). Solving this diff. eq we can find  $y$  (or  $x$ ) in terms of  $t$ . Sub in (1) or (2) we can find  $x$  (or  $y$ ) in terms of  $t$ .

Solve (i)  $\frac{dx}{dt} - y = t, \quad \frac{dy}{dt} + x = 1$

(ii)  $\frac{dx}{dt} + x = y + e^t, \quad \frac{dy}{dt} + y = x + e^t$

(i)  $Dx - y = t \rightarrow (1)$

$x + Dy = 1 \rightarrow (2)$

(1)  $\times D, \quad D^2x - Dy = D(t) \Rightarrow D^2x - Dy = 1 \rightarrow (3)$

(3)  $+ (2), \quad D^2x + x = 2 \Rightarrow (D^2 + 1)x = 2 \rightarrow (4)$

A.E is given by  $D^2 + 1 = 0, \quad D = \pm i$

CF =  $C_1 \cos t + C_2 \sin t$

PI =  $\frac{1}{D^2 + 1} (2) = \frac{1}{D^2 + 1} (2e^{0x}) = \frac{1}{0+1} (2) = 2$

Gen soln of (4),

$x = C_1 \cos t + C_2 \sin t + 2$

From eq (1)

$$y = Dx - t$$

$$= D[C_1 \cos t + C_2 \sin t + 2] - t$$

$$= -C_1 \sin t + C_2 \cos t - t$$

Required solution is given by

$$x = C_1 \cos t + C_2 \sin t + 2$$

$$y = -C_1 \sin t + C_2 \cos t - t$$

$$(ii) \quad \frac{dx}{dt} + x = y + e^t ; \quad \frac{dy}{dt} + y = x + e^t$$

$$\text{or } Dx + x - y = e^t, \quad Dy + y - x = e^t$$

$$\text{or } (D+1)x - y = e^t \rightarrow (1)$$

$$-x + (D+1)y = e^t \rightarrow (2)$$

$$(1) \times (D+1), \quad (D+1)^2 x - (D+1)y = (D+1)e^t \dots$$

$$\text{or } (D+1)^2 x - (D+1)y = 2e^t \rightarrow (3)$$

$$(2) + (3), \quad (D+1)^2 x - x = 3e^t$$

$$(D^2 + 2D + 1 - 1)x = 3e^t$$

$$\text{or } (D^2 + 2D)x = 3e^t \rightarrow (4)$$

$$\text{A.E. is } D^2 + 2D = 0, \quad D(D+2) = 0 \Rightarrow D = 0, -2$$

$$\text{CF} = C_1 e^{0t} + C_2 e^{-2t} = C_1 + C_2 e^{-2t}$$

$$\text{PI} = \frac{1}{D^2 + 2D} (3e^t) = \frac{1}{3} (3e^t) = e^t$$

$\therefore$  solution of eq (4) is given by

$$x = C_1 + C_2 e^{-2t} + e^t$$

$$y = (D+1)x - e^t$$

$$= (D+1)(C_1 + C_2 e^{-2t} + e^t) - e^t$$

$$= -2C_2 e^{-2t} + e^t + C_1 + C_2 e^{-2t} + e^t - e^t$$

$$y = C_1 - C_2 e^{-2t} + e^t$$

Required soln is given by

$$x = c_1 + c_2 e^{-2t} + e^t$$

$$y = c_1 - c_2 e^{-2t} + e^t$$

Solve (1)  $\frac{dx}{dt} - 7x + y = 0$ ,  $\frac{dy}{dt} - 2x - 5y = 0$

$$(D-7)x + y = 0, \quad (D-5)y - 2x = 0$$

$$(D-7)x + y = 0 \rightarrow (1)$$

$$-2x + (D-5)y = 0 \rightarrow (2)$$

$$(1) \times (D-5) \quad (D-5)(D-7)x + (D-5)y = 0 \rightarrow (3)$$

$$-2x + (D-5)y = 0 \rightarrow (2)$$

$$(3) - (2), \quad (D-5)(D-7)x + 2x = 0$$

$$(D^2 - 12D + 35 + 2)x = 0$$

$$\therefore (D^2 - 12D + 37)x = 0$$

$$\text{A.E.} \quad D^2 - 12D + 37 = 0$$

$$D = \frac{12 \pm \sqrt{144 - 148}}{2} = \frac{12 \pm \sqrt{-4}}{2}$$

$$= \frac{12 \pm 2i}{2} = 6 \pm i$$

$$x = e^{6t} [c_1 \cos t + c_2 \sin t]$$

From eq (1)  $y = -(D-7)x$   
 $= -(D-7) [e^{6t} (c_1 \cos t + c_2 \sin t)]$

$$\therefore y = - (e^{6t} (-c_1 \sin t + c_2 \cos t) + 7e^{6t} (c_1 \cos t + c_2 \sin t))$$

$$= e^{6t} [c_1 \sin t - c_2 \cos t + c_1 \cos t + c_2 \sin t]$$

$$= e^{6t} [(c_1 - c_2) \cos t + (c_1 + c_2) \sin t]$$

Solve (1)  $\frac{dx}{dt} + 4x + 3y = t$ ,  $\frac{dy}{dt} + 2x + 5y = e^t$

(2)  $\frac{dx}{dt} - y = t$ ,  $\frac{dy}{dt} + x = t^2$